

ASRI: A Structural Measure of Occupational Substitution and Reshaping—An Analytical Framework Beyond Automation Probability

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Core Exposition

The ASRI model can be understood in plain terms as a highly innovative index that is both intuitively comprehensible and clearly descriptive of the bidirectional tension between occupational substitution risk and reshaping dynamics brought about by AI development. Distinguished from the prevailing unidirectional exposure indices, ASRI adopts a substitution-defense bidirectional six-dimensional structure, underpinned by a reasonably justified scoring standard and formula weighting logic. Its complete expanded form is:

$$ASRI = [(Degree\ of\ Proceduralization \times 0.5) + (Degree\ of\ Digitalization \times 0.3) + (Verifiability\ of\ Outcomes \times 0.2)] \times 0.6 - [(Ambiguous\ Adaptability \times 0.35) + (Interpersonal\ Empathy \times 0.35) + (Non-standard\ Operations \times 0.3)] \times 0.4$$

Abstract

Against the backdrop of artificial intelligence rapidly permeating the world of work, discussions about "which occupations will be replaced" are increasingly prevalent. However, existing judgments often remain at the level of occupational labels, easily overlooking the significant differences in substitutability among different tasks within the same occupation. To address this issue, this paper proposes the AI Substitution and Reshaping Index (ASRI) V1.0, a composite index for measuring the structural substitution pressure an occupation faces in the era of AI. The core stance of ASRI is that AI's impact on occupations occurs first at the task level rather than the occupational label level. Therefore, its object of assessment is not the abstract occupational title, but the task composition structure of a specific occupation within a specific scenario.

ASRI V1.0 deconstructs occupational task structures into six dimensions, further grouped into two mutually countervailing forces. Among them, the Degree of Proceduralization, Degree of Digitalization, and Verifiability of Outcomes constitute the substitution dimension group; Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations constitute the defensive dimension group. On this basis, this paper constructs a three-tier formula system consisting of six-dimensional scoring, sub-index aggregation, and total index calculation. The substitution sub-index and defensive sub-index are each formed through weighted methods, ultimately yielding the composite indicator: $ASRI = (S \times 0.6) - (D \times 0.4)$. This structure aims to express the net structural substitution pressure between the parts of occupational tasks more susceptible to AI takeover and those still requiring human undertaking.

This paper further provides a foundational framework for an ASRI scoring manual, clarifying its scoring unit, dimensional definitions, continuous scoring principles, and case-based application methods. Through occupational scenario analysis, it demonstrates that ASRI is better suited to revealing intra-occupational task restructuring, capability differentiation, and structural reshaping, rather than simply predicting whether an occupation will disappear entirely. The paper also points out that ASRI is not an unemployment rate model, an elimination probability model, or a substitution timeline forecasting tool; nor is it used to judge occupational value. It is an explanatory framework serving structural analysis.

As a V1.0 version, ASRI has already formed a relatively complete conceptual definition, formula structure, and scoring logic. However, limitations still exist, including scoring subjectivity, yet-to-be-calibrated inter-rater reliability, a limited case database, and the fact that institutional friction costs have not been explicitly incorporated. Nevertheless, the proposal of ASRI still offers a more fine-grained and more interpretable task structure analysis pathway for understanding occupational changes in the AI era than the simple question of "will this occupation be replaced?". The significance of this paper lies in advancing the previously generalized discussion of occupational substitution into a

methodological framework that is decomposable, comparable, reviewable, and continuously calibratable.

Keywords

ASRI; AI Substitution and Reshaping Index; Artificial Intelligence; Occupational Substitution; Occupational Reshaping; Task Composition; Structural Substitution Pressure; Occupational Analysis

Introduction

Artificial intelligence is penetrating the world of work at an unprecedented pace. Whether in content generation, data processing, customer service, educational support, or programming, design, and management collaboration, AI is no longer merely an external tool but is increasingly embedding itself into the internal processes of occupational activities. This transformation has made "which occupations will be replaced" a high-frequency question in public discourse, industry judgment, and policy attention. However, a great deal of discussion surrounding this issue still generally remains at the level of macroscopic judgment based on occupational titles. For example, people often directly debate "whether teachers will be replaced," "whether lawyers will be replaced," or "whether designers will be replaced," as if an occupation itself were a fixed unit capable of withstanding the impact of artificial intelligence holistically, synchronously, and uniformly.

This mode of discussion has clear limitations. In reality, an occupation has never been a homogeneous whole but rather a composite structure composed of multiple tasks, responsibility relationships, interaction forms, and execution conditions. Under the same occupational title, different job scenarios often possess vastly different work emphases and task compositions; even within the same position, highly proceduralized and easily digitalizable parts frequently coexist with highly contextualized parts that rely on responsible judgment and non-standard execution. Consequently, the impact of AI on occupations rarely manifests as "the entire occupation disappearing wholesale," but more often as certain tasks being automated first, some tasks being augmented by AI, others being reallocated, and still others remaining in human hands. In other words, what AI first alters is often not the occupational label itself, but the task structure within the occupation.

If the holistic questioning mode of "will this occupation be replaced?" continues to be used, it easily obscures this more fine-grained process of change and readily misinterprets technological impact as a uniform judgment of outcomes. In fact, what many occupations experience in the AI era is not necessarily complete disappearance, but more likely the restructuring of internal tasks, the reordering of capability structures, the redrawing of job boundaries, and even significant differentiation within the same occupation. Therefore, it is necessary to propose a method capable of conducting structured analysis of occupations at

the task level, so as to more accurately describe the substitution pressure and reshaping trends that occupations face under AI conditions.

This paper proposes the AI Substitution and Reshaping Index (ASRI) V1.0 precisely based on this problem awareness. The fundamental stance of ASRI is that the impact of artificial intelligence on the world of work first occurs at the task level, not the occupational label level. Hence, the object of ASRI's analysis is not the abstract occupational name, but the **task composition structure of a specific occupation in a specific scenario**. The core question it attempts to answer is not "will a certain occupation disappear?" but rather "which tasks within that occupation are more susceptible to AI takeover, restructuring, or automation, and which tasks still constitute a defensive core?" In this sense, what ASRI measures is the **structural substitution pressure** borne by an occupation, rather than the unemployment outcomes, elimination probabilities, or substitution timelines that would inevitably occur in reality.

To achieve this goal, ASRI V1.0 deconstructs the occupational task structure into six dimensions and further organizes them into two sets of forces: one set comprises substitution factors that propel tasks toward greater susceptibility to AI takeover, and the other set comprises defensive factors that cause tasks to remain dependent on human undertaking. The former includes the Degree of Proceduralization, Degree of Digitalization, and Verifiability of Outcomes; the latter includes Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations. Through six-dimensional scoring, sub-index aggregation, and total index calculation, ASRI seeks to advance the discussion of occupational substitution from impressionistic judgment to a methodological framework that is decomposable, interpretable, comparable, and reviewable.

It should be noted that the proposal of ASRI is not intended to provide a deterministic model of the future fate of all occupations. This paper does not claim that occupational changes are determined solely by task structure, nor does it deny the significant role of external factors such as institutional arrangements, liability attribution, legal regulation, social trust, industry norms, cost structures, and the speed of technological diffusion. On the contrary, ASRI adopts a deliberately converging methodological strategy: it focuses first on the relatively analyzable layer of intra-occupational task structure, attempting to provide a structured "ruler" for the complex issue of occupational change. Only on this basis can subsequent research further discuss the relationship between institutional friction, industry differences, and realistic substitution pathways.

The work of this paper primarily encompasses the following aspects. First, it defines the conceptual connotation, analytical object, and methodological boundaries of ASRI, explaining why it takes task composition as its core rather than using occupational labels as the direct unit of assessment. Second, it proposes the six-dimensional structure and dual-group logic of ASRI V1.0, explaining how substitution factors and defensive factors jointly constitute the net structural substitution pressure of an occupation. Third, it presents the

corresponding formula system and the foundational framework of the scoring manual to ensure the basic operability of the model. Subsequently, through occupational scenario cases, it demonstrates the application value of ASRI in analyzing intra-occupational differences, explaining job restructuring, and identifying defensive cores. Finally, this paper will also explicitly discuss the limitations of ASRI V1.0, including issues such as scoring subjectivity, the not yet systematically calibrated inter-rater reliability, the need for further expansion of the case database, and the fact that institutional implementation layer variables have not been separately incorporated.

The significance of this paper does not lie in providing a simple ranking of "who will disappear first," but in proposing a new analytical perspective: rather than viewing occupations as static labels, it is better to understand them as dynamic structures composed of different tasks; rather than asking whether an occupation as a whole is safe, it is better to analyze which internal parts are bearing stronger structural substitution pressure. The proposal of ASRI is precisely intended to advance the generalized debate about the relationship between AI and occupations into a more fine-grained and more explanatory methodological discussion.

Compared with existing unidirectional automation risk assessments that take the occupation as the whole unit, ASRI's core advancement lies in two points: first, it takes task composition as the unit of analysis, realizing a paradigm shift from "occupational judgment" to "task diagnosis"; second, through its substitution-defense dual-group countervailing logic, it avoids reducing occupational change to a linear process driven unilaterally by technological capability. Within this framework, this paper particularly proposes the distinction between "performative empathy" and "responsible empathy," providing a more precise conceptual tool for identifying those interpersonal defensive tasks that cannot be simply replaced by AI.

In this sense, ASRI V1.0 is not a final model, but the starting point of a methodological framework. It attempts to provide a preliminary yet usable structural analysis language for understanding occupational changes in the AI era and to lay the foundation for subsequent case accumulation, scoring calibration, and empirical research.

Literature Review

1. The Task Model Tradition: Shifting the Unit of Analysis from Occupational Labels to Task Structures

Discussions on how technological progress affects the division of labor and occupational structures have long been accumulated in economics and sociology. However, the methodological breakthrough of shifting the unit of analysis from abstract occupational labels down to concrete work tasks began with the pioneering study by Autor, Levy, &

Murnane [1], *The Skill Content of Recent Technological Change: An Empirical Exploration*. They proposed that computer technology does not replace "high-skill" or "low-skill" labor, but rather replaces "routine tasks"—those work contents that can be described by explicit rules, steps, and procedures. This finding not only explained the polarization phenomenon in the U.S. labor market at the time, but more importantly, it established the task, rather than the occupation, as the fundamental unit of analysis for technological substitution. Acemoglu & Autor [2] further developed this task model framework in *Skills, Tasks and Technologies: Implications for Employment and Earnings*, categorizing tasks into multiple dimensions such as routine versus non-routine, and cognitive versus manual, providing a systematic analytical foundation for subsequent research.

The core insight of this tradition is that the impact of technology on labor first occurs at the task level, not the occupational label level. Any occupation is composed of several tasks of different natures, and the substitutability of these tasks in the face of technology is not uniform. The research of this paper is built precisely upon this methodological premise, and it seeks to make a further advancement on this basis: expanding task analysis from a single "routine/non-routine" dimension to a six-dimensional structural framework encompassing both substitution factors and defensive factors.

2. Occupational Substitution Assessment in the AI Era: Contributions and Limitations

With the rapid expansion of artificial intelligence, especially generative AI and large language models, the discussion on occupational substitution has once again been pushed to the forefront. Unlike early computer technology that mainly acted on rule-defined back-office tasks, current AI is beginning to permeate fields such as content generation, knowledge processing, interactive communication, and even creative collaboration. This has led to a significant increase in public anxiety and academic debate about "which occupations will be replaced."

Within this wave of discussion, the study by Frey & Osborne [3], *The Future of Employment: How Susceptible Are Jobs to Computerisation?*, has been the most influential. They assessed the computerization probability of 702 U.S. occupations and concluded that 47% of employment was at high risk. This research sparked widespread public discussion, but its methodology has a fundamental limitation: the unit of analysis is the occupation as a whole. They assigned a single "substitution probability" to each occupation, thereby assuming that "accountant" or "teacher" is a homogeneous whole, ignoring the significant differences in substitutability among different tasks within the same occupation. As Autor [4] pointed out in *Why Are There Still So Many Jobs? The History and Future of Workplace Automation*, occupational labels are often merely products of institutional classification, not the true units of technological substitution.

In recent years, some studies have attempted to advance the analysis to a more fine-grained level. Eloundou, Manning, Mishkin, & Rock [5], in *GPTs Are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models*, utilized the O*NET task database to assess the impact of large language models like GPT on over 19,000 specific tasks, proposing the concept of "exposure." Felten, Raj, & Seamans [6], in *Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses*, similarly constructed an AI occupational exposure index at the task level. These studies have improved analytical precision, yet they still essentially follow a unidirectional analytical logic: they measure "how much AI can do," rather than systematically examining "which tasks structurally require human undertaking." Furthermore, Huang & Rust [7], in *Artificial Intelligence in Service*, proposed four types of intelligence for AI to replace human labor from the perspective of service intelligence, providing a useful theoretical categorization for understanding the capability boundaries of AI. However, their framework likewise does not incorporate defensive factors into a countervailing structure against substitution factors.

Overall, existing research on AI occupational substitution has two main limitations. First, the unit of analysis often still remains at the occupational label level, and even when it descends to the task level, it predominantly relies on unidirectional assessments of substitutability or exposure. Second, the analytical perspective is mostly driven by technological capability, with less systematic discussion of the task structure characteristics within occupations that constitute a defensive core.

3. Research Gap and Positioning of This Paper

Based on the above review, this paper identifies two core gaps in current research. First, there is a lack of a bidirectional analytical framework capable of simultaneously characterizing both the substitution factors and the defensive factors of tasks. Existing studies either assess "substitutability" or measure "exposure," but none systematically analyze "which tasks are more susceptible to AI takeover" and "which tasks still require human undertaking" as countervailing forces within the same framework. Second, there is a lack of a meso-level structural analytical tool situated between macroeconomic employment statistics and micro-level demonstrations of technological capability, making the substitution pressure within occupations difficult to decompose, compare, and review.

This paper proposes the AI Substitution and Reshaping Index (ASRI) V1.0 precisely to fill the aforementioned gaps. Inheriting the task model tradition, ASRI explicitly sets the unit of analysis as the "task composition structure of a specific occupation in a specific scenario," rather than the abstract occupational name. On this basis, ASRI deconstructs the task structure into six dimensions—Degree of Proceduralization, Degree of Digitalization, Verifiability of Outcomes, Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations—and further organizes them into a substitution dimension group and a defensive dimension group. Through the countervailing calculation of these two sets of forces, what ASRI outputs is not a single "substitution probability," but the net structural

substitution pressure within the occupational task structure.

Within this framework, this paper particularly proposes the distinction between "performative empathy" and "responsible empathy." With the enhanced capabilities of generative AI in linguistic and emotional expression, the traditional concept of "emotional labor" is no longer sufficient to accurately identify the true sources of defensibility. ASRI posits that what constitutes the defensive core is not superficial emotional expression, but whether the interactant needs to bear actual responsibility for relational judgments and interaction outcomes. This distinction provides a more precise conceptual tool for analyzing interpersonal work in the AI era.

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Research Method

This study aims to develop an analytical framework capable of measuring the structural substitution pressure borne by occupations in the era of artificial intelligence from the task structure level. The core tasks of the research method are to propose the conceptual definition of ASRI, construct the six-dimensional scoring structure, establish the formula system, design the analytical framework, and formulate an operable scoring manual. This chapter will sequentially elaborate on ASRI's measurement framework, calculation logic, analytical pathway, and scoring implementation specifications.

1. The Measurement Framework of ASRI

The AI Substitution and Reshaping Index (ASRI) is a composite index used to measure the structural substitution pressure an occupation faces in the era of artificial intelligence. Its core concern is not to judge whether a particular occupation will disappear entirely, nor to rank occupations by their value, but to analyze, within the task composition of an occupation in a specific scenario, which parts are more susceptible to being substituted, restructured, or taken over by artificial intelligence, and which parts still need to be continuously undertaken by humans.

Therefore, the object measured by ASRI is not the abstract occupational label itself, but the **task composition structure** within the occupation. In other words, what ASRI cares about is not "what is this occupation called?", but "what tasks does this occupation primarily consist of in reality, and what structural characteristics do these tasks each possess?" This definition constitutes one of the most important differences between ASRI and general discussions of occupational substitution.

1.1 Object of Assessment: Task Composition Structure of a Specific Occupation in a Specific Scenario

The scoring object of ASRI is, in principle, not a generalized occupational title, but the **task composition structure of a specific occupation in a specific scenario**. This means that under the same occupational title, different positions, organizational environments, and work emphases may yield significantly different ASRI results.

For example, "teacher" is not a naturally unified scoring object. Compared to directly scoring "teacher" as a whole, objects more suitable for ASRI assessment are typically occupational scenarios with clearer task structures, such as "exam-oriented training teacher," "ordinary secondary school homeroom teacher," "university foundational course lecturer," or "special education teacher." Although they all fall under "teacher," they may exhibit significant differences in responsibility relationships, interpersonal load, degree of proceduralization, and task verifiability, and therefore should not be treated as the same scoring unit.

This also means that the occupational title serves merely as an analytical entry point in ASRI, while the task composition is the true unit of analysis. Only by deconstructing an occupation down to a sufficiently specific scenario level can ASRI accurately characterize its structural substitution pressure.

1.2 Methodological Stance: Task Composition over Occupational Labels

ASRI is built upon a fundamental judgment that the impact of artificial intelligence on the world of work first occurs at the **task level**, not the occupational label level. In reality, occupations are rarely substituted in a monolithic, synchronous, and uniform manner. More commonly, certain tasks are automated first, some are augmented by AI, others are reallocated, and still others remain in human hands.

Therefore, rather than directly asking "Will a certain occupation be replaced by AI?", it is better to further inquire: which tasks within that occupation are more susceptible to substitution, which are more prone to reshaping, and which still constitute its defensive core? The proposal of ASRI is precisely intended to systematically articulate this structural differentiation at the task level, moving the discussion—which previously remained at the level of intuition and impression—into an analytical framework that can be decomposed, compared, and reviewed.

1.3 Dual-Group Structural Logic

ASRI V1.0 deconstructs the occupational task structure into six dimensions, further grouping them into two mutually countervailing forces: one group embodies the tendency for tasks to be taken over, standardized, and proceduralized by AI; the other group embodies the tendency for tasks to remain dependent on deep human involvement. The former constitutes the substitution dimension group (S-Group), and the latter constitutes the defensive dimension group (D-Group).

The Substitution Dimension Group (S-Group) describes the parts of a job that are more susceptible to being standardized, automated, and procedurally taken over by artificial intelligence. It includes the following three dimensions: Degree of Proceduralization, Degree of Digitalization, and Verifiability of Outcomes. These three jointly constitute the substitution foundation within occupational tasks. Generally speaking, the more easily a task can be broken down into stable procedures, the more it primarily occurs in a digital environment, and the more it possesses clear, verifiable output standards, the more sufficient its structural conditions for AI takeover, embedding, or automated processing become.

The Defensive Dimension Group (D-Group) describes the parts of a job that are more difficult for AI to stably undertake and still require deep human involvement. It includes the following three dimensions: Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations. These three jointly constitute the defensive foundation within occupational tasks. Generally speaking, the more a task depends on making judgments in uncertain situations, on responsible complex interactions, and on non-standard execution in complex real-world environments, the higher the difficulty of it being stably substituted by AI in the short to medium term.

The core idea of ASRI is this: whether an occupation bears strong structural substitution pressure does not depend on its social prestige, educational threshold, or occupational sanctity, but on how substitution factors and defensive factors are distributed and countervail within its internal task structure.

1.4 Conceptual Definitions of the Six Dimensions

Degree of Proceduralization refers to whether a task possesses an execution structure that is relatively stable, repeatable, and decomposable into standard procedures. If the task steps are clear, exceptions are few, and the execution path is relatively fixed, its Degree of Proceduralization is high; conversely, if the task frequently requires departing from existing procedures for impromptu handling, its Degree of Proceduralization is low.

Degree of Digitalization refers to whether the input, processing, and output of a task primarily occur within digital systems, and whether its information can be recorded, transmitted, processed, and reorganized in digital form. If the work process is mainly completed within software, platforms, databases, or network systems, its Degree of Digitalization is high; if a large amount of work still relies on offline environments, unstructured physical contact, or on-site activities that are difficult to digitally capture, its Degree of Digitalization is low.

Verifiability of Outcomes refers to whether task outcomes can be easily inspected, compared, judged as right or wrong, or measured for quality based on explicit standards. If the work outcomes have stable answers, unified rules, clear indicators, or explicit acceptance criteria, its Verifiability is high; if the outcomes are highly dependent on contextual interpretation, long-term feedback, or make it difficult to form a unified judgment criterion, its Verifiability is low.

Ambiguous Adaptability refers to whether a task requires continuous judgment and on-the-spot adjustment under conditions of incomplete information, changing situations, insufficient rules, or conflicting goals. If the work frequently requires redefining problems, identifying exceptions, balancing conflicting goals, and immediately revising strategies under uncertain conditions, its Ambiguous Adaptability is high; if the task is mainly executed under conditions with clear rules and limited variables, its Ambiguous Adaptability is low.

Interpersonal Empathy refers to whether a task depends on trust-building, emotional processing, relationship maintenance, and contextualized understanding, and requires the practitioner to take responsibility for the consequences of the interaction. Its core is not just "whether communication with people is needed," but whether the interaction possesses deep responsibility. If the interpersonal interaction in the work directly affects judgment quality, relationship continuity, the state of the subject, or actual outcomes, its Interpersonal Empathy is high; if the interaction mostly remains at the level of superficial

expression and procedural contact, its Interpersonal Empathy is low.

Non-standard Operations refers to whether a task depends on complex, variable, and difficult-to-fully-script real-world operations, especially the execution capabilities requiring fine motor skills, hand-eye coordination, and on-site adjustment in physical environments. Its core is not whether "hands are used," but whether non-standardized operations must be completed under complex real-world conditions. If the work environment is changeable, execution conditions are unstable, and action paths are difficult to predefine, its Non-standard Operations score is high; if the operations are highly standardized, with fixed steps and easily scriptable, its Non-standard Operations score is low.

1.5 Key Distinction within Interpersonal Empathy: Performative Empathy vs. Responsible Empathy

Within the dimension of Interpersonal Empathy, ASRI V1.0 particularly emphasizes the distinction between "performative empathy" and "responsible empathy." This distinction is not a rhetorical detail, but is key to understanding why certain occupations, seemingly highly interpersonal on the surface, may still be partially substitutable structurally.

Performative empathy refers to displaying understanding, comfort, warmth, and companionship through language, tone, expression, or interaction style. This type of empathy has strong imitability, and under certain conditions, artificial intelligence can generate similar surface-level manifestations. Responsible empathy, in contrast, requires the practitioner not only to make an emotional response, but also to bear actual responsibility for the judgment behind the response, the ensuing relationship, and the consequences of the interaction. This responsibility may manifest as professional responsibility, legal liability, ethical responsibility, or responsibility within an ongoing relationship.

ASRI holds that what truly constitutes defensibility is not the superficial ability to express emotion, but the ability to bear the consequences of judgment within complex interpersonal relationships. Therefore, a high Interpersonal Empathy score is not equivalent to "being good at comforting people," but is closer to "whether one must be responsible for empathetic judgments."

1.6 Meaning of ASRI Results

The ASRI result is used to express the **net structural substitution pressure** after the combined effect of substitution factors and defensive factors within an occupational task structure. "Net" here refers to the comprehensive countervailing outcome of substitution and defensive factors within the same task structure, rather than the simple strength or weakness of a single dimension.

Generally, a higher ASRI result means that within the task structure of that occupational scenario, the parts more susceptible to AI takeover, restructuring, or automation are relatively stronger; a lower ASRI result means that the parts requiring sustained human undertaking—such as ambiguous judgment, responsible interaction, or non-standard execution—are relatively stronger within that occupation. If the result falls in the middle range, it typically indicates that the occupation has both significant substitutable parts and defensive parts, and is more likely to experience task restructuring, rather than simply disappearing entirely or remaining unchanged.

Therefore, what ASRI reflects is not a binary judgment of "will it disappear or not?", but a distribution of pressure on the task structure.

1.7 What ASRI Is Not

To avoid misunderstanding, it is necessary to further clarify which concepts ASRI should not be equated with.

First, ASRI is not an unemployment rate prediction model. It does not directly predict how many people in a certain occupation will lose their jobs. Second, ASRI is not an occupational elimination probability model; it does not represent the "probability of an occupation being wiped out." Third, ASRI is not a substitution timeline; it does not answer how many years it will take for an occupation to be replaced by AI. Simultaneously, ASRI is not a tool for ranking occupational value—a high score does not mean an occupation is inferior, nor does a low score mean it is superior. Finally, ASRI is not a tool for evaluating individual capability; it assesses the occupational task structure, not the competence level of an individual practitioner.

These distinctions are important because the methodological goal of ASRI from the outset is not to label occupations simply as "safe" or "dangerous," but to provide a more fine-grained analytical framework for understanding structural changes within occupations.

1.8 Theoretical Uses of ASRI

The primary use of ASRI V1.0 is to compare the task structures of different occupations or positions, analyze which parts within an occupation are more susceptible to substitution, explain whether an occupation is more likely to be "substituted" or "reshaped," and provide a more fine-grained analytical framework for discussions on labor structural change. Furthermore, ASRI can serve as a foundational indicator for subsequent case studies, occupational classification, methodological calibration, and cross-occupational comparison.

Its value lies not in providing a single conclusion, but in advancing the discussion of "AI's impact on occupations" from generalized judgment to structural analysis. It attempts to answer not "Can this occupation still exist?", but "Which tasks within this occupation will change first, which capabilities still constitute the defensive core, and does this change more closely resemble substitution or reshaping?"

1.9 Version Boundaries of V1.0

ASRI is currently at version V1.0. This version has already formed a relatively complete structural definition, clarifying the assessment object, the six-dimensional framework, and the dual-group logic of substitution and defensibility, and already possesses an operable scoring framework. Therefore, V1.0 is not merely a purely conceptual idea, but a theoretical scoring framework that can be practically applied.

However, V1.0 is still in the preliminary shaping stage of a theoretical model. Its subsequent refinement still depends on more case accumulation, scoring calibration, and consistency testing. It is also for this reason that V1.0 should be understood as a structural analysis framework that already possesses use-value but has yet to undergo large-scale empirical verification, rather than a closed, finished model requiring no further calibration.

2. Formula System

The formula system of ASRI V1.0 is designed to measure the **structural substitution pressure** an occupation faces in the era of artificial intelligence. Its object of analysis is not the occupational title itself, but which parts of that occupation's internal task structure are more easily taken over by AI, and which parts still constitute the defensive boundary of human labor.

Therefore, ASRI does not adopt a single total score or a simple equal-weight average of the six dimensions. Instead, it uses a "dual sub-index + net value formula" structure to respectively characterize the substitution forces and defensive forces within occupational tasks, and then calculates the net relationship between the two.

2.1 Three-Tier Calculation Structure

ASRI V1.0 employs a three-tier calculation structure.

The first tier is the input scoring for the six basic dimensions. For the same assessment object, each of the six dimensions is assigned a continuous score from 0.0 to 1.0: Degree of Proceduralization, Degree of Digitalization, Verifiability of Outcomes, Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations.

The second tier is the construction of two sub-indices. The six dimensions are not entered into the total formula directly and simultaneously; rather, they are first divided into two sub-indices based on the direction of their effects: the **Task Substitution Index (S)** and the **Irreplaceable Depth Index (D)**.

The third tier is the calculation of the final ASRI value. After obtaining S and D, the total formula is used to calculate the net structural substitution pressure for that occupation.

The purpose of this design is to avoid mixing "structural factors that are easier to substitute" with "structural factors that constitute human defensibility" into a non-directional total score.

2.2 Task Substitution Index S

The Task Substitution Index (S) measures the parts of an occupational task structure that are more readily copied, automated, standardized, and organizationally taken over by artificial intelligence. S is composed of three dimensions: Degree of Proceduralization, Degree of Digitalization, and Verifiability of Outcomes. Its calculation formula is as follows:

$$S = (\text{Degree of Proceduralization} \times 0.5) + (\text{Degree of Digitalization} \times 0.3) + (\text{Verifiability of Outcomes} \times 0.2)$$

In this structure, the Degree of Proceduralization is given the highest weight because whether a task can be stably broken down into clear, repeatable, and reusable steps typically constitutes the primary prerequisite for AI access and substitution. The Degree of Digitalization is weighted second; its role lies in determining whether the input, processing, and output of a task can more readily enter digital systems and be embedded and restructured by AI. The Verifiability of Outcomes mainly reinforces the stability and implementability of substitution; when task outcomes are easier to inspect, compare, and judge, the evaluation cost for an organization to adopt AI decreases significantly.

2.3 Irreplaceable Depth Index D

The Irreplaceable Depth Index (D) measures the parts of an occupational task structure that are more difficult for artificial intelligence to stably undertake and still require deep human involvement. D is composed of three dimensions: Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations. Its calculation formula is as follows:

$$D = (\text{Ambiguous Adaptability} \times 0.35) + (\text{Interpersonal Empathy} \times 0.35) + (\text{Non-standard Operations} \times 0.3)$$

Among them, Ambiguous Adaptability reflects whether the work requires the practitioner to continuously make re-judgments amid incomplete information, insufficient rules, conflicting goals, or changing situations. Interpersonal Empathy does not merely refer to superficial emotional soothing or gentle language, but emphasizes the responsibility-taking within the interaction—whether the practitioner needs to be responsible for communication outcomes, relational consequences, ethical risks, or psychological undertaking. Non-standard Operations primarily reflects whether the work requires completing fine, variable, and difficult-to-fully-script operations in complex real-world environments.

2.4 Total Formula

After calculating the Task Substitution Index S and the Irreplaceable Depth Index D , ASRI V1.0 employs the following total formula:

$$\text{ASRI} = (S \times 0.6) - (D \times 0.4)$$

This formula expresses two opposing forces within the occupational task structure: substitution factors drive the structural substitution pressure upward; defensive factors buffer the structural substitution pressure; the final ASRI result reflects the net value after the combined effect of both.

ASRI V1.0 does not use a fully symmetrical 0.5/0.5 structure, but adopts an asymmetrical configuration of 0.6/0.4. The reason is that what this research cares about is not the abstract total amount of "human uniqueness" in an occupation, but the structural substitution pressure borne by the occupation under the condition of AI diffusion. Under this problem awareness, task substitution is the active driving term for pressure formation, while irreplaceable depth manifests more as a buffering and decelerating term. Therefore, the total formula gives S a higher overall weight to reflect that substitutable structures often become the dominant variable of occupational impact earlier in the early stages of technological diffusion.

Meanwhile, ASRI V1.0 also does not adopt a stronger bias, such as 0.7/0.3. This is because an excessively strong one-sided configuration would compress the explanatory power of the defensive dimensions, underestimating the role of ambiguous judgment, responsible interaction, and non-standard real-world operations in the long-term process of occupational reshaping. 0.6/0.4 thus constitutes a median configuration: it neither treats substitution and defensibility as fully symmetrical, nor reduces defensibility to a nearly

ineffective ornamental term.

2.5 Why Not Use a Simple Six-Dimension Average?

ASRI V1.0 does not use a direct equal-weight average of the six dimensions, primarily for the following three reasons.

First, the directions of the six dimensions differ. The Degree of Proceduralization, Degree of Digitalization, and Verifiability of Outcomes mainly characterize the substitutable structures of tasks, while Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations mainly characterize the defensive structures of tasks. A direct average would smooth out the directional difference between the two categories of factors.

Second, the operating mechanisms of each dimension are not identical. Proceduralization, Digitalization, and Verifiability are not simply additive totals of an occupation's "advantages," but are structural conditions that make AI takeover easier; Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations are also not abstract totals of an occupation's "virtues," but structural sources that are harder to be stably substituted. The two should not be superimposed indiscriminately on the same level.

Third, a net value structure better aligns with ASRI's problem awareness. What ASRI cares about is not the overall complexity of an occupation, nor its overall value, but the tension relationship between substitution and defensibility within the occupation. Therefore, first constructing S and D, and then calculating the net value, better reflects the real structural pressure an occupation bears in the AI era.

2.6 Theoretical Range and Result Interpretation

Since all six basic dimensions adopt continuous scoring from 0.0 to 1.0, and both S and D are composed of weighted sums, the theoretical range of S is 0.0–1.0, and the theoretical range of D is 0.0–1.0. After substituting into the total formula, the theoretical range of ASRI is a minimum value of -0.4 and a maximum value of 0.6.

The meaning of this range can be understood as follows: when the result approaches -0.4, it indicates that the irreplaceable depth within the occupational task structure is significantly stronger than the task substitution; when the result approaches 0, it indicates an evident tug-of-war between substitution and defensibility, and the occupation is more likely to experience task reshaping, responsibility migration, or job redefinition; when the result approaches 0.6, it indicates that the substitutable parts within the occupational task structure significantly dominate, and the structural substitution pressure borne is relatively strong. Therefore, the ASRI result should be interpreted as the **net substitution pressure**

level within the occupational task structure, rather than a singular judgment on the future fate of a particular occupation.

2.7 Boundaries of Result Interpretation

It must be particularly pointed out that the ASRI result has clear interpretive boundaries. First, ASRI is not an unemployment rate; it does not directly predict the future unemployment scale of a particular occupation. Second, ASRI is not an occupational elimination probability; it does not provide a probability value for whether an occupation "will disappear." Third, ASRI is not a substitution timetable; it does not answer how many years it will take for an occupation to be replaced by AI. Fourth, ASRI is not a measure of individual capability; what it reflects is the occupational task structure, not whether a particular practitioner is good enough. Fifth, ASRI is not a ranking of social value; a high or low score does not constitute a judgment on occupational dignity, social significance, or ethical value.

2.8 Nature of the Weights

The weight configuration currently adopted by ASRI V1.0 belongs to **theoretically derived weights**. Its main bases include: structural analysis that AI substitution first occurs at the task level; judgments on the intensity of different dimensions' roles in substitution and defensibility; and the explanatory demand for real-world occupational differentiation phenomena. Therefore, the weights of V1.0 do not claim to have undergone large-scale empirical calibration. Future versions may further conduct empirical revision and statistical fitting of the weights through methods such as expert scoring, the Delphi method, large-sample occupational comparison, regression analysis, or cross-industry case validation.

3. Analytical Framework

The analytical framework of ASRI V1.0 aims to transform the issue of "how artificial intelligence affects occupations"—which typically remains at the level of macroscopic judgment or technological intuition—into a structured analytical process that can be decomposed, compared, interpreted, and reviewed. Its core task is not to directly judge whether a particular occupation will disappear entirely, but to identify, within the task structure of an occupation in a specific scenario, which parts are more susceptible to AI takeover, embedding, or procedural restructuring, and which parts still constitute the defensive core borne by humans, and to judge the net structural substitution pressure borne by that occupation accordingly.

From a methodological perspective, ASRI's analytical framework is built upon four levels: "unit of analysis—dimensional decomposition—dual-group aggregation—result interpretation." It does not use abstract occupational titles as the direct scoring object; instead, it first requires clearly defining the actual task scenario of the assessment object;

then decomposing that object into six dimensions for scoring; then aggregating the six dimensions according to the dual-group logic of substitution and defensibility; and finally performing structural interpretation of the obtained results. Thus, ASRI forms a complete analytical pathway from occupational scenario identification to structural pressure judgment.

3.1 Unit of Analysis: Specific Occupational Scenario as the Basic Object

The starting point of the ASRI analytical framework is the clear definition of the unit of analysis. Its basic unit is not the abstract occupational name, but the **task composition structure of a specific occupation in a specific scenario**. This determines that ASRI does not directly perform generalized scoring of broad occupational labels such as "teacher," "doctor," "lawyer," or "customer service," but requires further clarification of the actual job scenario, organizational environment, scope of responsibility, and task emphasis.

The reason is that under the same occupational title, task structures can differ enormously. Different organizations, business models, institutional responsibilities, and workflows can significantly alter a position's Degree of Proceduralization, Degree of Digitalization, interpersonal responsibility load, and non-standard execution requirements. If these differences are ignored and scoring is performed directly on the overall occupational title, it is easy to mistake highly internally heterogeneous positions for the same analytical object, thus weakening the explanatory power of the analysis results.

Therefore, the first step of ASRI is not scoring, but defining the object. An object suitable for ASRI analysis should generally possess the following characteristics: its work scenario is relatively clear, its task boundaries are basically describable, its main responsibility structure is relatively distinct, and its task emphasis can be distinguished from other scenarios of the same occupation. In other words, what ASRI requires is not that the occupational title is as broad as possible, but that the analytical object is as specific as possible.

3.2 Analytical Logic: From Task Decomposition to Structural Judgment

After clarifying the unit of analysis, ASRI enters the second level: task structure analysis. The core logic here is: AI's impact on occupations first occurs at the task level; therefore, the work content primarily undertaken in an occupational scenario needs to be understood as a combination of several tasks with different structural attributes, rather than an indivisible whole.

ASRI does not require infinitely detailed task-by-task coding for every occupation, but requires the scorer to always think around the **primary task composition** within that occupational scenario when making judgments. The basis for scoring is not the social impression of the occupational title, nor individual extreme cases, but the most

representative task structure characteristics of that position in reality. That is to say, what ASRI analyzes is not "what this occupation could theoretically do," but "what this occupation typically and mainly does in this specific scenario."

In this process, the scorer needs to continuously answer two questions. First, which parts of these tasks are more susceptible to AI standardization, proceduralization, digital takeover, and outcome verification? Second, which parts of these tasks still highly depend on human judgment, responsible interaction, and non-standard real-world execution in ambiguous situations? The former corresponds to substitution factors; the latter corresponds to defensive factors. The analytical logic of ASRI is precisely to simultaneously identify these two types of forces, avoiding the understanding of occupational change as a unidirectional technological push.

3.3 Six-Dimensional Decomposition: Scoring Based on Structural Attributes, Not Industry Impressions

In specific operation, ASRI decomposes the occupational task structure into six dimensions: Degree of Proceduralization, Degree of Digitalization, Verifiability of Outcomes, Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations. The setting of these six dimensions is not intended to exhaust all the complexity of an occupation, but to extract the key structural attributes that most influence AI takeover conditions and human defensive conditions.

This framework has an important feature: it assesses not the industry attributes of an occupation, nor its social status, but the structural characteristics of the tasks themselves. Therefore, high-education occupations are not naturally less substitutable, nor are low-education occupations naturally more substitutable; high-prestige occupations are not automatically safe, nor are low-prestige occupations necessarily more easily replaced. What ASRI always focuses on is the task structure: whether tasks are stably reproducible, whether they primarily occur in digital space, whether they are easily verifiable by uniform standards, whether they require adjustment in uncertain situations, whether they depend on responsible interactions, and whether they must be completed through non-standard execution in complex real-world environments.

This dimensionalization process allows different occupational scenarios to be compared under a relatively unified structural language. It does not require different occupations to do the same things, but requires that different occupations can all be placed under the same set of structural questions for analysis. Consequently, ASRI can not only explain differences between occupations, but also explain why different job scenarios under the same occupational title may exhibit different substitution pressures.

3.4 Dual-Group Aggregation: The Countervailing Structure of Substitution and Defensibility

Within the analytical framework, the six dimensions do not exist in isolation from each other but are further organized into two mutually countervailing forces. The Degree of Proceduralization, Degree of Digitalization, and Verifiability of Outcomes jointly constitute the substitution dimension group, representing the structural conditions that make tasks easier for AI to take over, embed, standardize, and automatically process; Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations jointly constitute the defensive dimension group, representing the structural sources that still require deep human involvement, responsibility-taking, and on-site execution.

The significance of this dual-group logic is that an occupation's substitution pressure is never determined by a single dimension. Even if a position has highly explicit procedures and highly digitized work, if its key links still highly depend on ambiguous judgment or responsible interaction, it may not necessarily exhibit extremely high net substitution pressure. Conversely, even if a position sounds very "professional" or "complex," if its core tasks actually mainly manifest as standardized information processing, rule invocation, and verifiable outputs, it may equally bear strong structural substitution pressure.

Therefore, ASRI does not understand occupational analysis as simply looking for "the most easily substitutable features," but as a structural countervailing between substitution factors and defensive factors. What the final result reflects is not how high a single dimension is individually, but the net pressure direction formed by the combined effect of the two sets of forces.

3.5 Result Interpretation: Moving from Numerical Judgment to Structural Explanation

The final step of the ASRI analytical framework is not to stop at the score itself, but to enter the structural explanation of the result. The score is merely a form of information compression; its true meaning lies in helping the analyst understand: which task parts within an occupational scenario are becoming more susceptible to AI takeover, which tasks still constitute the occupational defensive core, and what kind of change pathway the occupation is more likely to experience in the future.

Therefore, the analytical results of ASRI should not be understood as a final verdict on the fate of an occupation. A higher ASRI value does not equate to the inevitable disappearance of the occupation, but more likely means that a larger proportion of its task structure already meets the conditions for AI restructuring, embedding, and standardized takeover; a lower ASRI value does not equate to absolute occupational security, but more likely indicates that it still has a strong human defensive core, thus appearing more as assistive augmentation rather than complete substitution in the short to medium term. For

occupations in the middle range, the most typical scenario is often not a simple division of "safe" or "dangerous," but uneven changes occurring among different tasks within the occupation, leading to job function migration, capability requirement reordering, and hierarchical differentiation within the occupation.

In other words, the result of ASRI is better suited to answering "how will it change?" rather than "will it exist?". This point also marks an important difference between ASRI and automation probability-type discussions. It is not used to create a list of occupations to be eliminated, but to analyze the structural change pathways within occupations.

3.6 Application Pathway of the Analytical Framework

From a practical application perspective, the ASRI analytical framework can generally be unfolded along the following pathway: (1) Define the analytical object, clarifying the specific occupational scenario; (2) Identify the primary task composition, sorting out the core work content and responsibility structure; (3) Perform six-dimensional scoring, making continuous judgments on the task structure across the six dimensions; (4) Form the dual-group sub-indices, aggregating the six-dimensional scores into S and D respectively; (5) Calculate the total ASRI value; (6) Perform structural interpretation, analyzing whether the occupation is more likely to experience substitution, embedding, restructuring, or reshaping; (7) Conduct horizontal or longitudinal comparisons.

This application pathway illustrates that ASRI is not merely a static scoring table, but a complete analytical process from object definition to result interpretation. What is truly valuable is not "obtaining a number," but the structural analysis that unfolds around that number.

3.7 Boundaries of the Analytical Framework

It should be further pointed out that although the ASRI analytical framework attempts to characterize the substitution pressure within occupational task structures, its focus remains limited to the structural level, not the institutional implementation level. That is to say, what ASRI answers is "from a task structure perspective, is an occupation more susceptible to AI takeover or restructuring?", not "when, at what speed, and under what institutional conditions will it be substituted in the real world?"

The real-world substitution process is still influenced by external factors such as liability attribution, legal regulation, industry norms, organizational adoption, user trust, technological costs, and social acceptance. Therefore, the analytical results of ASRI should be understood as a structural a priori judgment, not a direct prediction of real-world outcomes. It provides a pressure direction, not a fate timetable; it is a structural explanation, not a real-world statistical conclusion.

This boundary does not weaken the significance of ASRI; rather, it clarifies its mode of use as an analytical tool. Only by distinguishing the task structure layer from the institutional implementation layer can we avoid confusing technological capability judgments with real-world substitution processes and leave space for subsequent research that integrates institutional variables for deeper investigation.

4. Scoring Manual

To ensure the operability, reviewability, and methodological transparency of ASRI V1.0 across different occupational cases, this paper further proposes a foundational scoring manual. This manual standardizes the definition of scoring objects, the six-dimensional scoring method, the scoring process, recording requirements, and consistency control principles. The core question this manual answers is: **How should ASRI be practically scored in specific occupational scenarios?**

4.1 Scoring Objects and Basic Principles

The scoring object of ASRI V1.0 is not the abstract occupational label, but the **task composition structure of a specific occupation in a specific scenario**. The implementation of scoring should adhere to the following basic principles:

Scenario-First Principle: Before scoring, the specific work scenario of the occupation should first be defined, followed by the six-dimensional judgment. Scoring directly on a macroscopic occupational title detached from its scenario can easily obscure significant differences in intra-occupational task structures, thereby reducing assessment precision.

Task Composition Principle: The basis for scoring should be built upon the primary task structure within that occupational scenario, not on social impressions, occupational halos, educational thresholds, industry myths, or subjective imaginings of individual capability. What ASRI cares about is how tasks are organized, executed, and verified, not what social prestige the occupational label itself holds.

Structural Judgment Principle: What ASRI assesses is the substitution pressure and defensive strength of a task structure in the face of artificial intelligence, not the social value, occupational dignity, occupational meaning, or individual level of the practitioner. A higher ASRI for an occupation does not mean its social significance is lower; a lower ASRI does not mean its labor is nobler or more advanced.

Comprehensive Judgment Principle: The scoring for each dimension should be based on a

comprehensive judgment of the overall characteristics of the primary task combination within that scenario, rather than seizing upon a marginal task, special case, or extreme situation to determine the entire score based on a single impression.

Explainability Principle: Each dimensional score must be accompanied by a brief written justification. A score without a scoring rationale should not be considered a complete scoring result. ASRI emphasizes the traceability of results, so each score value must be explainable by returning to the task structure level.

Uniform Scale Principle: All six dimensions adopt a unified continuous interval of 0.0–1.0, but this score value represents the "strength of that dimension within that task structure," not a measure of "goodness or badness" or "superiority or inferiority."

Prudent Interpretation Principle: The result of ASRI expresses structural substitution pressure and should not be directly interpreted as an unemployment rate, occupational elimination probability, occupational value ranking, or explicit substitution timeline. It is a structural analytical indicator, not a fate-prediction tool.

4.2 Method for Defining the Scoring Object

To avoid misjudgment caused by overly coarse occupational titles, ASRI recommends writing the scoring object in the following format: **Occupational Title + Scenario Delimitation + Task Emphasis Description**. For example: E-commerce Customer Service (Standardized After-Sales Inquiry Scenario), Secondary School Homeroom Teacher (Scenario Combining Classroom Teaching and Student Management), 3D Modeler (Game Asset Mass Production Pipeline Scenario), 3D Modeler (High-End Film and Television Character Customization Scenario), etc.

When significantly different task scenarios exist within the same occupation, they should be split into multiple scoring objects and scored separately, rather than forcibly merged. Before formal scoring, the scorer should complete the object delimitation description, including at least the following four items: occupational title, scenario boundary, list of core tasks, and marginal tasks not included in this scoring.

If the scoring object's boundaries are unclear, tasks are ambiguous, or the scenario is indefinite, it should not proceed to formal scoring. Meanwhile, scoring should, as much as possible, be based on the **typical task structure** within that scenario, not on the high-level tasks undertaken by a few top practitioners, nor on idealized task pictures from occupational promotions.

4.3 Scoring Scale and Recording Rules

ASRI V1.0 adopts **0.0–1.0 continuous scoring**, rather than integer-tiered scoring. This continuous scale helps to more finely express the gradient differences in occupational task structures. Generally, 0.0 indicates extremely low or almost non-existent, around 0.25 indicates low, around 0.50 indicates medium, around 0.75 indicates high, and 1.0 indicates extremely high or highly typical.

Decimals can be used in scoring, such as 0.20, 0.35, 0.60, 0.72, 0.85, etc., but it is not recommended to use overly finely divided decimals to create a "sense of precision." The basic version of scoring recommends controlling precision at the level of **0.05 or 0.10** to balance differentiation and reviewability. If the scorer cannot clearly explain the difference between 0.55 and 0.60, then forced subdivision should not be applied.

4.4 Six-Dimensional Scoring Anchors

ASRI V1.0 sets scoring anchors for each of the six dimensions to improve relative consistency among scorers, but these do not constitute a mechanical scoring template. The final score value should still return to the specific task composition and scenario boundary for judgment. The scoring anchors for the six dimensions are detailed in the Appendix.

4.5 Standard Scoring Procedure

The standard scoring procedure for ASRI V1.0 consists of six steps: (1) Define the scoring object, clarifying the occupational title, scenario scope, core tasks, and tasks not included; (2) List the primary task composition, recommended to be summarized in 3–8 items; (3) Score dimension by dimension, assigning a 0.0–1.0 score to each of the six dimensions with a brief justification; (4) Calculate S, D, and ASRI; (5) Write the result interpretation, explaining the sources of substitution pressure, sources of defensibility, the change pathway the occupation is more likely to experience, and the key scenario assumptions; (6) Conduct a review, checking for issues such as whether the occupational label itself was used as the basis for scoring, whether the scoring is inconsistent with the textual explanation, or whether the scenario delimitation was too coarse.

4.6 Scoring Record Template

To facilitate case accumulation and review, it is recommended to use a unified record format for each scoring object. A basic record template can include: scoring object information (occupational title, scenario delimitation, task emphasis description, scoring date, scorer, data sources), primary task composition, tasks not included, six-dimensional scoring table (score and justification for each dimension), result calculation (S, D, ASRI), and a brief interpretation.

4.7 Scoring Subjectivity and Consistency Control

ASRI V1.0 is currently still a theory-driven structural scoring tool, thus inevitably containing a degree of scoring subjectivity. ASRI should not pretend to achieve absolute objectivity by "eliminating subjectivity," but should strive to improve scoring consistency and discussability through clear principles, anchor references, scoring justifications, and review mechanisms. The basic version recommends the following control methods: require each dimensional score to be accompanied by a brief justification; base scores as much as possible on public information, task descriptions, and scenario facts; adopt independent scoring by multiple persons for comparison in important cases; retain documentation of disagreements for cases with significant differences rather than forcing unification; and gradually introduce inter-rater consistency testing, expert calibration, or Delphi method revisions in future versions.

In this sense, scoring divergence itself does not necessarily mean model failure; on the contrary, it often exposes problems such as unclear delimitation of occupational scenario boundaries, insufficient task description, or not yet fully unified understanding of dimensions. Therefore, retaining necessary disagreements and explaining their sources is part of the methodological transparency of ASRI V1.0.

4.8 Common Misconceptions and Applicability Boundaries

In practical use, several typical misconceptions easily arise in the ASRI scoring process and should be avoided: directly using the occupational title as the scoring object while neglecting specific scenario differences; replacing task structure analysis with occupational prestige, educational thresholds, or social impressions; mistaking the high-level tasks of a few top practitioners for the universal tasks of that occupation; directly equating "very busy at work" or "great responsibility" with high Ambiguous Adaptability or high Interpersonal Empathy; misjudging "frequent communication with people" as high responsible empathy; simplistically equating a high Degree of Digitalization with inevitable high substitution; and mechanically interpreting the ASRI total score as some kind of unemployment probability or occupational life-or-death verdict.

It should also be noted that ASRI V1.0 currently does not advocate mechanically dividing the total score into several fixed tiers. Rather than pursuing rigid thresholds, it is better to understand the source of the result by combining the internal structures of S and D, the specific task composition, and the scenario assumptions. ASRI is better suited to identifying the formation mode of structural pressure, rather than replacing specific industry research, labor market statistics, or individual occupational diagnosis.

Occupational Cases

Case 1: Psychological Counselor (Ongoing Individual Counseling Scenario)

Scoring Object

Psychological Counselor (Private Practice or Institutional Ongoing Individual Psychological Counseling Scenario)

Scenario Description

The psychological counselor in this case provides individual psychological counseling services in a private practice studio or a professional psychological institution, with sessions held once a week, each lasting 50–60 minutes, continuing for several months or more. The counselor faces clients experiencing emotional distress, interpersonal difficulties, traumatic experiences, or psychological disorders. The core of the work is to establish a therapeutic trust relationship through dialogue, conduct diagnostic assessment and intervention judgment under conditions of perpetually incomplete information, and bear ethical and legal responsibility for the consequences of interventions. The counselor must continuously handle clients' emotional outbursts, silent resistance, relational tensions, and crisis risks (such as self-harm or suicide risk). This scenario does not include single-session psychological hotlines, psychoeducational lectures, or corporate EAP group counseling—these forms of work differ significantly from this case in the depth of interpersonal interaction and the nature of responsibility.

Primary Task Composition

- Establish and maintain a therapeutic trust relationship through weekly sessions.
- Continuously assess the client's psychological state, risk level, and core conflicts under conditions of incomplete information.
- Flexibly adjust intervention strategies and conversational direction based on the client's immediate state.
- Handle intense emotional reactions (crying, anger, dissociation, silence) that arise during sessions.
- Identify and respond to crisis situations (self-harm, suicide risk, abuse reports), initiating protective interventions and referral procedures when necessary.
- Bear professional responsibility for the content of each session and intervention decisions, and maintain compliant clinical records.

Tasks Not Included in This Scoring

- Psychoeducational lectures and community mental health promotion—fall under educational dissemination, with significantly different interpersonal depth.

- Corporate EAP group counseling—falls under group work, with lower trust relationships and responsibility depth than individual counseling.
- Single-session psychological hotlines or initial reception—though involving emotional processing, do not form an ongoing responsible relationship.
- Clinical case research and academic writing—fall under cognitive work, not direct clinical tasks.

Six-Dimensional Scores and Justifications

Degree of Proceduralization: 0.10

The direction of each session cannot be predetermined. Although theoretical orientations (such as psychodynamic, cognitive-behavioral, humanistic, etc.) provide reference frameworks, in each specific session, the counselor must adjust in real time based on the client's state that day, the content expressed, emotional fluctuations, and relational tension. There is no SOP that can be "followed step-by-step"; the process varies greatly among different clients, and different stages with the same client may also require completely different approaches. Highly proceduralized psychological interventions—such as standardized CBT self-help programs—are, by definition, different service types from the ongoing in-depth counseling assessed in this case.

Degree of Digitalization: 0.25

The core work occurs in face-to-face interpersonal interaction within the counseling room. Although the counselor may use digital tools to record case notes or conduct remote video counseling, diagnostic information is primarily obtained through nonverbal cues (facial expressions, body posture, tonal shifts, the length and texture of silences) and the relational dynamics emerging from the interaction—information that is currently still difficult for digital systems to fully capture and encode. While video counseling increases the proportion of digitalization, the core interpersonal perception and interaction quality have not been changed by it.

Verifiability of Outcomes: 0.05

The effectiveness of psychological intervention is highly dependent on the client's subjective report, behavioral observation, and long-term follow-up, and is difficult to test against unified standards of "right/wrong" or "meeting/not meeting criteria." The same intervention may have vastly different effects on different clients; improvement in the same client may also be attributed to life events outside of counseling, social support, or natural recovery. Although the profession has assessment tools (such as symptom scales), interpreting these data itself requires professional judgment, and different theoretical orientations have philosophical disagreements on the definition of "effective." Identifying errors is extremely difficult—a seemingly ineffective session might be the prelude to a breakthrough, and a seemingly smooth session might mask avoidance and compliance.

Ambiguous Adaptability: 0.90

This is the core state of a psychological counselor's daily work. The information presented by the client is perpetually incomplete—they will forget, avoid, embellish, and unconsciously conceal. The causal chain is highly opaque—a symptom may be the common product of multiple conflicts, and intervention effects may not manifest until weeks later. Persistent conflicts exist among multiple goals—pushing for exploration vs. maintaining stability, challenging defenses vs. respecting pacing, protecting life safety vs. respecting client autonomy. The counselor must continuously make judgments in every session, unable to rely on fixed rules. Experience provides a better basis for judgment, but does not eliminate judgment itself.

Interpersonal Empathy: 0.95

This is a quintessential representative scenario of ASRI's concept of "responsible empathy." The counselor not only needs to understand the client's emotional state and psychological situation, but also bears full responsibility for the accuracy of that understanding, the appropriateness of the response, and the safety of the intervention. When a client expresses self-harm ideation, the counselor cannot merely say "I understand you are in great pain," but must make professional judgments—has the risk level reached the point of needing to breach confidentiality? Does an emergency contact need to be activated? Should a psychiatric referral be recommended? These judgments will directly affect another person's life safety, and the consequences are entirely borne by the counselor. This is the key to ASRI's distinction between "performative empathy" and "responsible empathy": AI can simulate the linguistic posture of empathy, but cannot bear responsibility for the consequences of empathetic judgments.

Non-standard Operations: 0.10

Psychological counseling involves almost no non-standard operations in the physical environment. The physical presence of the counselor has therapeutic significance, but this "presence" is different from the fine motor skills and hand-eye coordination involved in physical operations. This dimension is scored low.

Calculation Results

$$S = (0.10 \times 0.5) + (0.25 \times 0.3) + (0.05 \times 0.2) = 0.135$$

$$D = (0.90 \times 0.35) + (0.95 \times 0.35) + (0.10 \times 0.3) = 0.678$$

$$ASRI = (0.135 \times 0.6) - (0.678 \times 0.4) = -0.190$$

Case Explanation

The ASRI result for this case is -0.190, falling in the significantly negative region of the theoretical range, indicating that defensive factors overwhelmingly dominate substitution factors within the task structure. Compared to the CS2 professional player (ASRI = -0.079), the psychological counselor's defensibility is stronger, and the structural source of this defensibility is entirely different—the CS2 professional player relies on a combination of Non-standard Operations and Ambiguous Adaptability, while the psychological counselor relies on a combination of Interpersonal Empathy and Ambiguous Adaptability.

This structural difference is significant. It demonstrates that "low substitution pressure" is not a single category, but can be achieved through different combinations of defensive dimensions. The low ASRI of the CS2 professional player and the low ASRI of the psychological counselor have different sources and structures, and thus their coping pathways when facing technological impact are also different. ASRI's six-dimensional structure allows this difference to be tracked and discussed, rather than being compressed into a single "low risk" label.

The most central theoretical contribution of this case lies in providing a complete empirical anchoring for the concept of "responsible empathy." Against the backdrop of rapidly developing generative AI, AI is already capable of simulating surface-level empathetic behaviors such as understanding, comforting, and companionship at the linguistic level. The defensive core of the psychological counselor does not lie in "whether one can speak empathetically," but in "whether one can bear the consequences for empathetic judgments." When a client is at risk of self-harm, the counselor must make a judgment that could affect a life—a judgment that cannot be undertaken by AI, not only because of AI's insufficient accuracy, but more importantly because responsibility cannot be outsourced to a non-human entity at the institutional, ethical, and legal levels. By incorporating "responsible empathy" into the core connotation of the Interpersonal Empathy dimension, ASRI provides an analytical tool for identifying this type of structural human defensibility.

Primary Sources of Substitution Pressure: The substitution dimensions are overall at a low level. Although the Degree of Digitalization (0.25) is slightly above zero, it is insufficient to constitute a significant condition for AI access, because core clinical information is still primarily generated in face-to-face interpersonal interaction.

Primary Sources of Defensibility: The extremely high Interpersonal Empathy (0.95) and extremely high Ambiguous Adaptability (0.90) jointly constitute a powerful defensive structure. The defensive core lies not in the "appearance of empathy," but in the "responsibility for empathy."

Changes This Occupational Scenario Is More Likely to Experience: In the short to medium term, it is more likely to experience AI-assisted embedding (such as case note automation, preliminary assessment tools), rather than substitution of the core therapeutic relationship.

The core defensive tasks of the occupation—responsible empathy and ambiguous situational judgment—will still be retained in the hands of human practitioners. Low-complexity, standardized psychological services (such as self-service CBT) may experience more apparent substitution, but these are different task scenarios.

Key Scenario Assumptions: This scoring is strictly limited to ongoing in-depth individual counseling. If the scenario switches to standardized online psychological self-help courses or AI-assisted single-session psychological support, the Degree of Digitalization would significantly rise, Interpersonal Empathy might decrease due to a reduction in depth of responsibility, and the ASRI result would be significantly different. This comparison further illustrates that "task structure differences between different scenarios under the same occupational title" is the fundamental reason why ASRI must adhere to the scenario-first principle.

Case 2: CS2 Professional Player (S-Tier International Tournament Competitive Scenario)

Scoring Object

CS2 Professional Player (S-Tier International Tournament Competitive Scenario)

Scenario Description

The CS2 professional player in this case is contracted to a top-tier esports club, with the primary work scenario being participation in formal matches within S-Tier international tournaments (such as PGL Major, IEM Katowice, BLAST Premier, etc.). Matches are conducted in a closed LAN environment. The player operates the game character through personal peripherals, coordinating with four teammates to complete offensive and defensive tasks. Match outcomes are determined in real time by the game engine, with wins and losses accumulated on a per-map/per-match basis. This scenario does not include peripheral work such as daily training, streaming, or commercial activities—although these tasks occupy time in the professional player's career, they do not fall within the scoring scope of this case.

Primary Task Composition

- Complete high-precision shooting actions such as aiming, positioning, and spray control within millisecond time windows.

- Process multi-source battlefield information in real time, such as the minimap, footsteps, gunshots, and teammate callouts.
- Conduct immediate tactical communication and information transfer with teammates.
- Make on-the-fly judgments and strategic adjustments based on opponent tactics, economic status, and positional distribution.
- Maintain operational precision and decision-making quality in high-pressure decisive rounds.

Tasks Not Included in This Scoring

- Daily training (aim training, map practice, tactical drills)—Although training occupies a large amount of time, its task structure is fundamentally different from formal matches: the training environment lacks high-pressure confrontation, has no unpredictable opponent behavior, and produces no external consequences.
- Personal streaming and commercial activities—fall under the categories of content creation and public relations, not core competitive tasks.
- Tactical theory study and VOD review—fall under preparatory cognitive work, different from real-time decision execution in matches.

Six-Dimensional Scores and Justifications

Degree of Proceduralization: 0.15

There are established tactical frameworks in matches (such as default positions, pre-set utility combinations, common offensive routes), but these are merely pre-match setups; actual execution is highly dependent on on-the-fly changes. Every unconventional action by the opponent can lead to the immediate abandonment and reconstruction of the tactical framework. There is no task path that can be "completed simply by following an SOP." Precisely for this reason, coaches cannot control player behavior during matches through pre-set commands and must rely on the players' autonomous judgment.

Degree of Digitalization: 0.95

The input (game visuals, sound, teammate voice comms), processing (operating peripheral input commands), and output (in-game character actions) of the task are entirely completed within digital systems. All of the player's operational information is recorded, transmitted, and processed in digital form. The match server environment and peripheral drivers constitute a complete digital workspace.

Verifiability of Outcomes: 0.10

The final outcome of the match is determined by the game engine, which is verifiable at that level. However, the "quality" of an individual player's contribution is difficult to inspect against unified standards: a high kill count might indicate outstanding individual ability, or it might reflect team resource allocation favoring that player; a failed tactical execution might

stem from a player's operational error, or from the opponent's precise read. Competitive performance evaluation is highly dependent on the subjective interpretation of the coaching staff and analysts, with significant disagreements often existing among different evaluators. A player's "skill level" cannot be compared item-by-item against a source file like data entry results.

Ambiguous Adaptability: 0.95

This is where the core value of an esports professional player lies. Information during a match is perpetually incomplete—the opponent's position, intentions, economic status, and tactical choices can only be inferred through fragmented clues. Time pressure is extreme—engagement decisions are often made within 200 milliseconds. Persistent conflicts exist among goals—save equipment or take a risk for a kill, hold a position or push forward proactively, follow the tactic or adapt on the fly. The player must make high-quality judgments and execute them instantly under conditions of incomplete information, extremely constrained time, and persistent goal conflict. There are no fixed rules to rely on—the opponent is simultaneously adapting and countering.

Interpersonal Empathy: 0.40

Teamwork requires trust and coordination. Real-time information transfer among teammates (callouts, commands, emotional management) constitutes an important component of the task. However, this type of communication primarily functions as information exchange and tactical coordination, not requiring the practitioner to bear deep emotional responsibility or relational consequences. When morale is low, there is usually a specific player responsible for emotional regulation within the team (often the in-game leader or a veteran), but this responsibility is not evenly distributed among every player, and its depth is lower than that of typical responsible empathy positions such as psychological counselors or homeroom teachers.

Non-standard Operations: 0.98

This is the most prominent defensive dimension in this case. The aiming and spray control operations of a CS2 professional player involve extremely fine hand-eye coordination, muscle memory, and real-time micro-adjustments. Different weapons have different recoil patterns, different distances require different positioning strategies, and different opponent movement styles require different prediction paths. These operations cannot be fully scripted—not because the technology is insufficiently advanced, but because the operations themselves must be continuously adjusted in real-time feedback to opponent behavior. The trajectory of mouse movement, the rhythm and force of keystrokes, the subtle switching of perspectives—the precision and situational dependency of these operations constitute an extremely strong defensive structure. Unlike highly standardized keyboard data entry, CS2 operations are "a bodily response made after reading the opponent's intentions."

Calculation Results

$$S = (0.15 \times 0.5) + (0.95 \times 0.3) + (0.10 \times 0.2) = 0.380$$

$$D = (0.95 \times 0.35) + (0.40 \times 0.35) + (0.98 \times 0.3) = 0.767$$

$$ASRI = (0.380 \times 0.6) - (0.767 \times 0.4) = -0.079$$

Case Explanation

The ASRI result for this case is negative, indicating that defensive factors are significantly stronger than substitution factors within the task structure. This is one of the most prominent findings within this case database: the work of a CS2 professional player **occurs entirely within digital systems**, yet its structural substitution pressure is extremely low.

This result challenges a widespread but analytically crude intuition—that a high Degree of Digitalization necessarily implies high substitution pressure. The CS2 professional player's Degree of Digitalization is 0.95, equal to that of the Data Entry Clerk (0.95) among all cases, yet their ASRI results lie at opposite ends of the theoretical range: the Data Entry Clerk at 0.509 (extremely high substitution pressure), and the CS2 professional player at -0.079 (defensibility dominant). The source of the difference lies not in the Degree of Digitalization, but in the extreme difference in two defensive dimensions—Ambiguous Adaptability and Non-standard Operations. The Data Entry Clerk approaches zero on these two dimensions, while the CS2 professional player approaches full marks.

This comparison reveals a core proposition of the ASRI methodology: **The Degree of Digitalization is a condition for AI access, but not a sufficient condition for substitution to occur.** Whether substitution constitutes real pressure also depends on the proportion and intensity of defensive tasks within the occupational structure.

Furthermore, this case also illustrates that the dimension of "Non-standard Operations" still holds critical explanatory power in the digital age. The defensibility of Non-standard Operations exists not only in physical environments (such as the manual operations of a cleaner) but also in highly digitized competitive scenarios—when operations need to be continuously fine-tuned under millisecond time constraints and in response to unpredictable changes in opponent behavior, they form a defensive core that is difficult to simply script away.

Primary Sources of Substitution Pressure: The extremely high Degree of Digitalization provides AI with access conditions at the information input and processing level; the moderately high S value (0.380) suggests that the task substitution of this occupation is not

entirely negligible.

Primary Sources of Defensibility: The extremely high Non-standard Operations (0.98) and Ambiguous Adaptability (0.95) jointly constitute an overwhelming defensive structure. The core value lies in "making millisecond-level bodily and judgmental responses in an unpredictable confrontational environment."

Changes This Occupational Scenario Is More Likely to Experience: In the short to medium term, it is more likely to experience AI-assisted augmentation (tactical analysis, training optimization) rather than complete substitution. The core competitive tasks of the occupation remain in human hands, while peripheral tasks may be restructured by technology.

Key Scenario Assumptions: This scoring is strictly limited to the top-tier tournament competitive scenario. If the scenario expands to regular player matches or unranked matchmaking, the unpredictability of opponent behavior decreases, and Ambiguous Adaptability may be somewhat reduced, but Non-standard Operations still constitute the core defense. If the scenario switches to "game streamer," the task structure fundamentally changes—the core shifts from competitive operation to content production and audience interaction, and the ASRI result would be entirely different. This comparison itself illustrates that ASRI assessment must always adhere to the scenario-first principle.

Case 3: Web Novel Author (Paid Platform Daily Serialization Scenario)

Scoring Object

Web Novel Author (Paid Reading Platform Daily Serialization Scenario)

Scenario Description

The web novel author in this case serializes daily on paid reading platforms (such as Qidian Chinese Network, Tomato Novel, etc.), earning income through reader paid subscriptions, tipping, and platform revenue sharing. The author must maintain a stable daily update rhythm (typically 4,000–10,000 words per day), uphold the consistency of world-building and character settings during continuous output, and simultaneously adjust plot direction based on reader feedback data (subscription numbers, retention rates, comment content, recommendation tickets). The author also needs to manage reader communities (such as

QQ groups, WeChat groups), maintain fan relationships, and compete for recommendation slots and exposure resources under platform rules. This scenario does not include traditional literary creation completed in full before publication, nor does it include pure ghostwriting or writing studio order-based writing—these scenarios differ fundamentally from this case in terms of reader feedback pressure, daily update rhythm constraints, and platform ecosystem interaction.

Primary Task Composition

- Produce a prescribed word count of updated content daily, maintaining update frequency.
- Conceptualize and long-term maintain world-building architecture, character relationship networks, and main plot direction.
- Judge reader preferences based on subscription data, comment feedback, and recommendation ticket fluctuations, and adjust plot direction or character settings accordingly.
- Maintain fan relationships within reader communities, responding to reader expectations or controversies.
- Manage chapter pacing (cliffhangers, climax placement, suspense setup) to sustain paid conversion and retention rates.

Tasks Not Included in This Scoring

- IP adaptation negotiation and contract signing—falls under the commercial brokerage category, led by the contracted platform or agent.
- Cover design, typesetting, and platform backend operations—fall under technical support work.
- Omni-channel marketing and personal brand management—not a core creative task, and the difference on this item between top-tier and mid-tier authors on the platform is enormous.
- Ghostwriting or writing studio order-based writing—different scenario, completely different task structure, should be evaluated separately.

Six-Dimensional Scores and Justifications

Degree of Proceduralization: 0.30

The daily update rhythm forms a stable output cycle and fixed workload. Some genre novels have highly mature formulaic templates (such as "underdog counterattack," "broken engagement arc," "system narrative"), which provide reusable execution frameworks for specific plot nodes. However, core creation—world-building construction, character growth arc design, multi-plotline convergence—remains highly non-standardized; there is no path where "following an SOP can produce an excellent story." Proceduralization is manifested in the output rhythm and genre formulas, not in the core creative links.

Degree of Digitalization: 0.90

Writing, publishing, reader data acquisition, and reader interaction are all completed within digital platforms. Input (word processing software or writing platform), processing (text creation and revision), and output (chapter publishing) are fully digitized. Reader feedback—subscription numbers, retention rates, comments, monthly tickets—is entirely presented in structured data form in the author's backend. This is an occupational scenario highly embedded in the digital ecosystem.

Verifiability of Outcomes: 0.60

The market performance of web novels has clear quantitative metrics: subscription numbers, retention rates, recommendation tickets, monthly ticket counts, tipping amounts, and the work's ranking on platform charts. These data provide clear verification standards for judging "whether this work is popular." However, at the same time, these metrics measure market performance, not the "quality" of the text itself—literary aesthetics, narrative innovation, and character depth cannot be fully standardized and inspected. Therefore, Verifiability is at a moderately high level: commercial success is verifiable, artistic value is not.

Ambiguous Adaptability: 0.70

The web novel author needs to continuously make judgments and balance among multiple goals: the conflict between maintaining daily output pressure and pursuing text quality; the tension between conforming to reader preferences and adhering to personal creative intent; the contradiction between seeking short-term tipping revenue and building long-term world-building integrity. Although reader feedback data provides quantifiable reference, these data do not tell the author "how to write"—whether to follow the data and modify the plot, or trust one's own judgment and stick to the original direction; this decision itself requires judgment under highly ambiguous circumstances. Moreover, reader comments and community disputes are often mixed with misunderstandings, flame-baiting, and emotional venting; how to interpret this feedback, which parts to adopt and which to ignore, still depends on the author's implicit judgment.

Interpersonal Empathy: 0.65

The interpersonal interaction tasks of a web novel author are divided into two layers. The first layer is eliciting emotional resonance from readers through character creation—this requires the author to understand human emotions, desires, fears, and contradictions, and encode them into compelling characters and plots. The second layer is maintaining fan relationships through community interaction—responding to reader controversies over the plot, managing conflicts within the fan group, and providing emotional appeasement after hiatuses or plot controversies. Both layers of interaction go beyond "polite communication." When the author decides whether to alter a character's fate for fan preferences, this decision will affect the emotional experience and willingness to pay of a large number of readers, constituting an interaction with a certain depth of responsibility. However,

compared to a psychological counselor, the relationship between author and readers is not an ongoing individual responsible relationship, but a "one-to-many" content-emotional contract; the depth of responsible empathy is significantly lower than that of a counselor who must bear responsibility for the self-harm risk of a specific client.

Non-standard Operations: 0.05

The work of a web novel author is purely digital and cognitive labor, involving no non-standard operations in a physical environment. High proficiency in keyboard typing does not fall under the category of non-standard execution for this dimension.

Calculation Results

$$S = (0.30 \times 0.5) + (0.90 \times 0.3) + (0.60 \times 0.2) = 0.540$$

$$D = (0.70 \times 0.35) + (0.65 \times 0.35) + (0.05 \times 0.3) = 0.488$$

$$ASRI = (0.540 \times 0.6) - (0.488 \times 0.4) = 0.129$$

Case Explanation

The ASRI result for this case is 0.129, positioned in the mid-to-positive region of the theoretical range, indicating that substitution factors and defensive factors exhibit a clear tug-of-war dynamic within the task structure. This is the most complex case within this case database—its ASRI value is neither high nor low, and it is precisely this "medium" level that reveals the richest structural information.

The primary revelation of this case is: **"Creative work" is not inherently safe.** Public intuition often regards "writer" as an occupation difficult for AI to replace, but ASRI's decomposition reveals a more complex picture. The web novel author's substitution dimension group ($S=0.540$) is actually higher than the defensive dimension group ($D=0.488$); substitution pressure is not zero. Its substitution pressure primarily comes from two sources: an extremely high Degree of Digitalization (writing and publishing are entirely completed on digital platforms; AI already possesses access conditions at the text generation level), and a moderately high Verifiability of Outcomes (market data provides clear feedback signals, allowing "what constitutes a successful chapter" to be quantitatively evaluated). At the same time, defensibility comes from a combination of Ambiguous Adaptability and Interpersonal Empathy—making creative judgments amid multi-goal conflicts, and establishing emotional contracts with readers through characters and community interaction. The difference between the two is very small ($S-D = 0.052$), indicating that substitution and defensibility are almost evenly balanced.

The occupational change pathway pointed to by this structure is not the binary outcome of "being replaced" or "not being replaced," but **deep task restructuring and internal differentiation**. Specifically:

- **High Substitution Pressure Parts:** Tasks such as formulaic writing, chapter data monitoring and feedback analysis, and format standardization already structurally meet the conditions for AI embedding and assistance. AI can generate chapters that fit genre templates, recommend plot directions based on reader data, and imitate specific styles to complete filler content.
- **High Defensive Parts:** Original world-building construction, cross-chapter narrative architecture, ongoing emotional contract management with the reader community, and the judgment of "whether to abandon one's creative intent for the sake of data" remain the structural defensive core of human authors.

This means that what the occupation of web novel author is most likely to experience in the AI era is not overall disappearance, but the **intensification of internal capability stratification**: those authors whose core value is concentrated in formulaic writing and data analysis will face stronger substitution pressure, while those whose core value is built upon original world-building, deep narrative design, and reader emotional connection will still have an effective defensive core.

Horizontal comparison of this case with previous cases can further illustrate the analytical power of ASRI. **Different dimensional combinations can converge to similar ASRI values, yet point to entirely different restructuring pathways.** What ASRI provides is not just a total score, but a diagnostic structure that allows the restructuring direction to be traced back.

Primary Sources of Substitution Pressure: The extremely high Degree of Digitalization (0.90) opens AI access at the text generation level; the moderately high Verifiability of Outcomes (0.60) provides a quantitative standard for market feedback, making the optimization direction algorithmically drivable.

Primary Sources of Defensibility: The moderately high Ambiguous Adaptability (0.70) and Interpersonal Empathy (0.65) jointly constitute the defensive core. The key lies in creative judgment amid multi-goal conflicts, and the maintenance of the emotional contract with readers.

Changes This Occupational Scenario Is More Likely to Experience: Deep task restructuring and intra-occupational differentiation, rather than overall substitution. Formulaic writing may be assisted or substituted by AI; original architecture and emotional connection remain

the human defensive core. Clearer capability stratification may emerge within the occupation—writers adept at template-based production face higher substitution pressure, while those strong in world-building and reader empathy have stronger defensibility.

Key Scenario Assumptions: This scoring is strictly limited to the paid platform daily serialization scenario. If the scenario switches to pure literary free creation (no daily update pressure, no reader data feedback, publication cycle measured in years), Verifiability of Outcomes would drop significantly, and the Degree of Proceduralization might also decrease due to increased freedom in creative pacing, causing the ASRI result to shift markedly downward. Conversely, if the scenario switches to writing studio order-based writing following an outline, the Degree of Proceduralization and Verifiability of Outcomes would be higher, shifting the ASRI result upward. These differences further confirm that ASRI must adhere to the scenario-first principle—under the same occupational title, differences in task structure between different scenarios can be even greater than the differences between different occupations.

Case 4: Data Entry Clerk (Third-Party Data Service Company Batch Entry and Verification Scenario)

Scoring Object

Data Entry Clerk (Third-Party Data Service Company Batch Entry and Verification Scenario)

Scenario Description

The data entry clerk in this case is employed by a third-party data service company, providing outsourced data entry services for banks, insurance companies, or government agencies. The work content involves entering information from paper forms, scanned documents, or unstructured information into databases according to pre-set templates, and completing basic format verification. The clerk is managed by daily quotas; the operational process is strictly constrained by the company's standardized system; quality is verified through a dual acceptance process of automatic system comparison and manual spot-checking. This position typically does not require a professional background and can be started after short-term training.

Primary Task Composition

- Enter information from paper documents or scanned files into the database field by field.

- Verify data format, numerical range, and logical consistency according to established rules.
- Annotate and report data that is missing, ambiguous, or clearly anomalous.
- Meet the prescribed daily entry quota.
- Occasionally check unclear information with upstream data providers.

Tasks Not Included in This Scoring

- Database structure design and maintenance (handled by the technical department).
- Development and testing of new entry templates (handled by the project manager and technical department).
- Client communication and requirements alignment (handled by the account manager).

Six-Dimensional Scores and Justifications

Degree of Proceduralization: 0.90

The entry operation has an extremely stable sequence of steps: open template → read source file → enter data field by field → execute automatic validation → submit. The system itself is a mandatory SOP execution environment; the operational path is highly fixed, and exceptions are rare. New employees can work independently after just a few days of training.

Degree of Digitalization: 0.95

The input (scanned documents, electronic forms), processing (database entry software), and output (structured database records) of the task are entirely completed within digital systems. The work process can be fully recorded as an operation log; there is no meaningful offline component.

Verifiability of Outcomes: 0.85

Entry results can be compared with the source files field by field; format compliance is automatically verified by the system. Entry error rates can be precisely quantified, and quality acceptance standards are clear and explicit. Although occasional human judgment is needed for ambiguous information, the proportion is extremely low and can be handled through a rule manual.

Ambiguous Adaptability: 0.15

The vast majority of operations are strictly executed according to templates and essentially involve no independent judgment under conditions of incomplete information. Occasionally, situations such as illegible handwriting or abnormal formatting in source files occur, but the handling method typically follows fixed escalation rules (annotate and submit to a superior

or return to upstream), and does not constitute a substantial judgment load.

Interpersonal Empathy: 0.05

Daily work requires almost no meaningful interpersonal interaction. Occasional communication with upstream data providers is limited to information confirmation and does not involve trust-building, emotional processing, or relationship maintenance. The position can be entirely task-assigned and delivered through a work order system; interpersonal contact is not a necessary component.

Non-standard Operations: 0.05

The task is purely digital processing and involves no non-standard operations in a physical environment. High proficiency in keyboard and mouse operation does not fall under the category of non-standard execution for this dimension.

Calculation Results

$$S = (0.90 \times 0.5) + (0.95 \times 0.3) + (0.85 \times 0.2) = 0.905$$

$$D = (0.15 \times 0.35) + (0.05 \times 0.35) + (0.05 \times 0.3) = 0.085$$

$$ASRI = (0.905 \times 0.6) - (0.085 \times 0.4) = 0.509$$

Case Explanation

This case demonstrates substitution pressure levels approaching the extreme of the ASRI spectrum. The three substitution dimensions—Degree of Proceduralization, Degree of Digitalization, and Verifiability of Outcomes—are all in high-score ranges, meaning that structurally, virtually all core tasks of this position already meet the conditions for AI or automated system takeover. Meanwhile, the three defensive dimensions are all at extremely low levels, forming virtually no effective buffer.

It is noteworthy that the ASRI result for the Data Entry Clerk is not determined by its social prestige or educational threshold, but is directly derived from the structural characteristics of its tasks: highly proceduralized, highly digitized, and highly verifiable. This precisely illustrates the core stance of ASRI: the source of substitution pressure lies in the task structure, not in the social impression of the occupational label.

Horizontal comparison of this case with the CS2 professional player can more clearly reveal

this logic. Both occupations have a Degree of Digitalization of 0.95, but their ASRI results lie at opposite ends of the theoretical range: the Data Entry Clerk at 0.509 (extremely high substitution pressure), and the CS2 professional player at -0.079 (defensibility dominant). The root of the difference lies in the extreme divergence of defensive dimensions—the Data Entry Clerk approaches zero on all three dimensions of Ambiguous Adaptability (0.15), Interpersonal Empathy (0.05), and Non-standard Operations (0.05), while the CS2 professional player approaches full marks on Ambiguous Adaptability (0.95) and Non-standard Operations (0.98). This comparison confirms the bidirectional logic of ASRI: whether substitution occurs depends not only on how easily tasks can be taken over by AI, but also on whether structural defensive forces exist within the tasks.

From the perspective of occupational change pathways, this position is more likely to experience overall substitution or large-scale compression rather than task restructuring. In reality, this trend is already observable in industries where OCR and RPA technologies are widely applied.

Primary Sources of Substitution Pressure: Extremely high Proceduralization, Digitalization, and Verifiability of Outcomes jointly constitute substitution conditions with virtually no buffer.

Primary Sources of Defensibility: There is almost no meaningful defensive task structure.

Changes This Occupational Scenario Is More Likely to Experience: Overall substitution or large-scale compression of the position.

Key Scenario Assumptions: This scoring is based on the standardized outsourced entry scenario. If the entry content involves highly specialized medical or legal documents, the Verifiability of Outcomes might increase due to the need for professional judgment, thereby further intensifying substitution pressure; if a small amount of complex judgment is involved, Ambiguous Adaptability might rise slightly, but the impact on the overall ASRI direction for this scenario is limited.

Summary of Four Cases

Case	S	D	ASRI	Core Structural Characteristics
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Case	S	D	ASRI	Core Structural Characteristics
Psychological Counselor	0.135	0.678	-0.190	Extreme defense dominated by responsible empathy + ambiguous adaptability
CS2 Professional Player	0.380	0.767	-0.079	Digital high-defense dominated by non-standard operations + ambiguous adaptability
Web Novel Author	0.540	0.488	0.129	Substitution-defense tug-of-war, heterogeneous internal tasks
Data Entry Clerk	0.905	0.085	0.509	Extreme substitution with triple-high substitution + triple-low defense

Discussion

The proposal of ASRI V1.0 does not mean that the question regarding artificial intelligence and occupational change has received a final answer. On the contrary, it is more of a methodological repositioning: rather than continuing to cycle through the generalized debate of "will a certain occupation be replaced?", it is better to shift attention to the differences in intra-occupational task structures, analyzing how substitution factors and defensive factors jointly constitute the net structural substitution pressure borne by an occupation. In this sense, the significance of ASRI lies not only in providing a score, but in offering a new observational perspective, a new set of analytical language, and a methodological framework that can be continuously calibrated and expanded.

1. Shifting from Discussion Based on Occupational Labels to Discussion Based on Task Structure

One of the most central theoretical advancements of this paper lies in shifting the basic unit of occupational analysis down from abstract occupational labels to the "task composition structure of a specific occupation in a specific scenario." This shift holds significant importance. For a long time, public discussion and some research have been accustomed to directly comparing occupational titles themselves—such as teacher, lawyer, designer, programmer, customer service representative—and then judging whether these occupations "will be replaced by AI." However, this approach tends to treat occupations as internally homogeneous analytical objects with fixed boundaries, thereby overlooking intra-occupational task differences, organizational differences, and scenario differences.

The task structure perspective emphasized by ASRI implies that different distributions of substitution pressure may inherently exist under the same occupational title. The future of an occupation does not depend on the title itself, but on what tasks constitute its interior, what structural characteristics these tasks possess, which parts are more easily taken over

by AI, and which parts still require human undertaking. Such an analytical approach helps avoid premature conclusions and also allows occupational change to be understood as a process of internal restructuring, rather than a simple binary judgment of the survival or demise of an occupational label.

Compared to the automation probability assessment that takes the occupation as the whole unit of analysis, as seen in Frey & Osborne (2017), the methodological advantage of ASRI lies in this: it does not score "teacher" or "lawyer" as a whole, but requires specifying concrete task compositions such as "secondary school homeroom teacher (scenario combining classroom teaching and student management)" or "legal assistant (contract review and document collation scenario)." It is this scenario-based decomposition that enables ASRI to explain why there may be significant differences in the substitution pressure of different positions under the same occupational title—an explanatory power that the occupational label approach struggles to provide. The case studies also confirm this point: for occupations that are both highly embedded in digital environments, the CS2 professional player (ASRI = -0.079) and the Data Entry Clerk (ASRI = 0.509) have substitution pressures at opposite ends of the theoretical range. The root cause lies in the difference in defensive dimension structures, rather than the Degree of Digitalization itself.

2. Beyond Substitution, Reshaping is More Important

Although ASRI is named with "substitution," its more explanatory power often manifests in "reshaping." Real occupational changes, especially under the conditions of AI diffusion, rarely manifest as the neat and uniform disappearance of entire occupations. More commonly, low-complexity tasks are automated first, mid-level execution tasks are compressed, and tasks requiring high responsibility, high judgment, and high integration are elevated. Positions shift from output-oriented types to verification, supervision, collaboration, or integration types.

Therefore, the results of ASRI should not be simplistically understood as "high-scoring occupations are bound to disappear, low-scoring occupations are bound to be safe." A more reasonable understanding is that a high ASRI result means a larger proportion of the task structure in that occupational scenario already meets the conditions for AI embedding, takeover, standardization, or procedural restructuring; a low ASRI result means that the occupation still possesses relatively strong defensive structures, especially in terms of ambiguous judgment, responsible interaction, and non-standard real-world execution. The vast middle ground between the two may correspond precisely to the most common occupational form of the future: not complete substitution, but continuous reshaping.

The Web Novel Author case (ASRI = 0.129) most clearly demonstrates this logic. In this case, the S value (0.540) and D value (0.488) are almost equal; neither substitution nor defensive factors hold an overwhelming advantage in the task structure. What this structure points to is not the binary outcome of "will writers be replaced?", but deep task restructuring:

formulaic writing already structurally meets the conditions for AI embedding, while original world-building and the management of the emotional contract with readers still constitute the human defensive core. Capability stratification will emerge within the occupation—authors whose core value lies in template-based production will face higher substitution pressure, while those strong in original architecture and reader empathy will have stronger defensibility.

From this perspective, the value of ASRI is not just to describe "risk," but more importantly, to help identify exactly which types of tasks within an occupation will change first, and how job functions will migrate after those changes. This explanatory power is closer to reality than a simple "substitution or no substitution" binary judgment.

3. Why a Dual-Group Logic is Needed Rather Than a Unidirectional Substitutability Judgment

In existing discussions on AI and occupations, a common tendency is to only emphasize "which jobs are easily substituted," with less systematic discussion of "which parts constitute the defensive core of the occupation." If analysis proceeds only from substitutability, occupational analysis can easily slide into unidirectional technological determinism, explaining all occupational differences merely as temporary delays caused by the boundaries of technological capability, and struggling to explain why certain tasks structurally require sustained human undertaking.

The significance of ASRI dividing the six dimensions into a substitution dimension group and a defensive dimension group lies precisely in making it clear that occupational tasks have not only a "being substituted" side but also a structural source of "resisting substitution." Proceduralization, Digitalization, and Verifiability of Outcomes do indeed enhance the conditions for tasks to be taken over by AI; but meanwhile, Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations constitute another relatively stable set of human defensive sources. The net structural substitution pressure of an occupation is formed precisely through the countervailing of these two sets of forces.

This forms a key distinction from the unidirectional assessment logic centered on "exposure," as seen in Eloundou et al. (2023). Exposure assessment measures "how much AI can do," but differs from asking "why certain exposed tasks have still not been substituted in reality." By explicitly incorporating defensive factors into the same framework, ASRI makes this question analytically tractable. The CS2 professional player's high Degree of Digitalization (0.95) implies high exposure, but their extremely high Non-standard Operations (0.98) and Ambiguous Adaptability (0.95) form a structural hedge, resulting in a negative ASRI value—a phenomenon precisely unexplainable by unidirectional assessment.

This allows ASRI, compared to simple "automation risk" judgments, to better present the internal tension of occupational structures. The reason certain occupations do not disappear rapidly is not because they completely lack substitutable parts, but because their key defensive tasks are still sufficient to reconstitute job value. Conversely, certain high-education, high-status occupations are not necessarily inherently safe; if a large portion of their core tasks are structurally highly proceduralized, digitized, and verifiable, they may equally bear significant substitution pressure.

4. The Special Significance of "Responsible Empathy" as a Defensive Dimension

Among the six dimensions, Interpersonal Empathy deserves particular further discussion. The development of generative artificial intelligence means that "being able to talk," "being able to comfort," and "being able to simulate care" no longer naturally constitute exclusive human advantages. If defensibility is still understood through the traditional concept of emotional labor, the human irreplaceability of certain occupations can easily be overestimated. Therefore, ASRI particularly emphasizes the distinction between "performative empathy" and "responsible empathy."

This treatment has theoretical necessity. Many interpersonal interactions appear complex in form, but if their core is merely expressing politeness, soothing emotions, or generating a sense of companionship, then AI can already, at the surface level of performance, approach or even surpass many standardized interpersonal jobs to a certain degree. What truly constitutes defensibility is not whether the surface of the interaction is gentle or emotionally colored, but whether the interactant needs to take responsibility for relational judgments, subsequent impacts, and actual consequences. That is to say, the key to defensibility lies not in "what was said," but in "whether one must be responsible for what was said."

This point is particularly helpful in differentiating the quality of interpersonal tasks within different occupations. For example, positions such as hotline customer service, platform companionship, and standard consultation reception may have high interaction frequency, but do not necessarily all possess high responsible empathy; whereas positions such as psychotherapy, crisis intervention, classroom management, and complex negotiation—even if the linguistic form is not always "gentle expression"—may retain stronger defensibility precisely because they bear high degrees of responsible consequences. This distinction helps avoid treating all jobs that "deal with people" as inherently safe by default.

The Psychological Counselor case (ASRI = -0.190) provides complete empirical anchoring for this distinction. In this case, Interpersonal Empathy is as high as 0.95, but what the ASRI scoring relies on is not "the counselor is good at comforting people," but rather that when a client reveals self-harm ideation, the counselor must bear full professional and legal

responsibility for judgments on risk assessment, breaching confidentiality, emergency referral, etc.—AI can simulate the linguistic posture of empathy, but cannot be responsible for the consequences of empathetic judgments.

5. A Gap Still Exists Between Structural Pressure and Real-World Substitution

Although ASRI attempts to describe the structural pressure that occupations face from AI more precisely, structural pressure does not equate to the immediate occurrence of substitution outcomes in reality. Even if an occupation scores high on ASRI, it does not necessarily mean its real-world positions will disappear on a large scale simultaneously; conversely, certain mid-scoring occupations might experience significant restructuring earlier due to factors such as cost, organizational choices, or platform logic.

This reminds us that what ASRI characterizes is the substitutable pressure at the task structure level, not the implemented outcomes at the institutional realization level. In reality, whether technology can be transformed into job substitution depends on a large number of external variables, including whether regulatory systems permit it, whether organizations are willing to restructure processes, how responsibility is allocated, whether social trust is sufficient, whether the cost of error is bearable, and whether human users are willing to accept non-human executors. Particularly in industries with high responsibility, high risk, and high trust dependency, the institutional implementation layer often significantly delays the transformation of structural pressure into real-world substitution. For example, the ASRI of the psychological counselor is extremely low; this conclusion itself already hints that even if AI could technically simulate empathetic language, the ethical, legal, and institutional barriers involved in the counselor's responsible empathy would constitute additional obstacles to substitution beyond the structural pressure.

Therefore, ASRI is better suited as an "a priori structural ruler" to help identify which parts within an occupation are theoretically more easily taken over by AI, rather than serving as a direct timetable for the real-world substitution process. It provides a basis for understanding the direction of change, but cannot alone determine the speed of change.

6. The Explanatory Power of ASRI for Intra-Occupational Capability Differentiation

Another important value of ASRI is that it is not only applicable to comparisons between occupations, but also helps explain capability differentiation within the same occupation. With the diffusion of artificial intelligence, many occupations will not "rise as a whole" or "fall as a whole" internally; instead, clear stratification is more likely to emerge: low-complexity, standardizable, digitally takeoverable parts are compressed, while higher-order judgment, integration, relational responsibility, and complex execution capabilities are

elevated. This means that the pressure faced by different practitioners within the same occupation is not uniform.

This dimension-level traceability is an important feature that distinguishes ASRI from black-box models that output a single total score or probability value. By reporting S and D values separately, ASRI enables the analyst to trace the specific sources of substitution pressure—is the Degree of Proceduralization too high, or are the defensive dimensions too low? Is Ambiguous Adaptability insufficient, or is responsible empathy lacking? For the individual practitioner, this diagnostic capability can be directly translated into a reference for the direction of skill investment; for enterprises and policymakers, it can provide more precise entry points for job redesign and training programs.

In this context, ASRI is not only an analytical tool at the occupational level, but can also be understood as a tool for identifying the rearrangement of job functions. It helps explain why "junior positions" in certain occupations are impacted earlier than "senior positions," and also helps illustrate why some seemingly identically-named positions will rapidly embark on different developmental paths in the AI era. In other words, what ASRI provides is not just a horizontal comparison between occupations, but also a way of observing the vertical evolution of intra-occupational structures.

7. The Framework Significance and Version-Based Limitations of V1.0

It should be emphasized that the value of ASRI V1.0 does not lie in having completed all theoretical and empirical work on the issue of occupational substitution, but in having proposed a structural analysis framework that is sufficiently clear, decomposable, comparable, and reviewable. Compared to many discussions that remain at the level of impressionistic judgment, the advancement of ASRI lies in this: it transforms "occupational substitution risk" from a generalized judgment into an analytical object that can be located to specific dimensions, specific tasks, and specific scenarios.

Of course, this also means that V1.0 inevitably carries version-based limitations.

The Issue of Scoring Subjectivity. Although ASRI has attempted to minimize arbitrariness through measures such as clarifying the assessment object, adopting a six-dimensional structure, setting dimensional anchors, requiring written justifications, and adhering to the scenario-first principle, the scoring itself still inevitably contains judgmental components. During the methodological derivation for the "3D Modeler (Game Asset Mass Production Pipeline Scenario)," two independent scoring results differed by 0.084, indicating that scoring subjectivity is not an abstract concern but an observable reality already present in trial runs. Currently, ASRI V1.0 has not yet completed systematic multi-rater reliability

testing; this work is crucial for subsequent versions.

The Risk of "False Precision" in Continuous Scoring. The 0.0–1.0 continuous scoring provides higher resolution, but if scorers lack training, scores may appear precise while the justifications are insufficient to support the distinction, potentially leading to a "false precision" problem. True stability depends on scoring training, anchor consensus, and multi-rater review mechanisms.

Dimensional Boundary Adhesion. In actual scoring, empirical overlap or boundary adhesion may occur between certain dimensions, such as between the Degree of Proceduralization and Verifiability of Outcomes, between Ambiguous Adaptability and Interpersonal Empathy, and between the Degree of Digitalization and Degree of Proceduralization. Although ASRI has established an operable six-dimensional framework, the discriminant boundaries between dimensions still need further refinement through more case accumulation and scoring discussions.

The Operationalization Challenge of Responsible Empathy. ASRI has theoretically distinguished between "performative empathy" and "responsible empathy," but how to stably distinguish the two in practice remains a challenge. Some occupations appear "good at comforting people" yet may not bear sustained consequential responsibility. More case anchors are needed in the future to train scorers to identify the difference between the two.

The Limit of Temporal Robustness. The six dimensions of ASRI are built around relatively stable task structure attributes and possess a degree of temporal robustness, meaning they will not immediately become invalid due to the short-term capability fluctuations of a particular AI model generation. However, this temporal robustness does not mean ASRI can be directly applied indefinitely, detached from historical context—the ruler is relatively stable, but the object being measured changes. With technological diffusion and organizational process restructuring, the task structure of an occupational scenario itself may be redefined.

Institutional Friction Costs Not Yet Explicitly Incorporated. ASRI currently primarily models the task structure layer, not the institutional implementation layer. Factors such as the non-outsourcability of legal liability, excessive ethical risk, insufficient social trust, and regulatory systems not yet giving the green light can significantly delay the transformation of structural pressure into real-world substitution. In certain high-responsibility industries, the structural pressure expressed by ASRI may precede real-world substitution.

The Case Database Still Needs Expansion. Currently, four cases cover the ASRI range from - 0.190 to 0.509, but the number of cases remains limited. More types of occupational scenarios need to be covered in the future to enhance anchor stability and cross-industry

comparability.

These limitations do not negate the methodological significance of ASRI. On the contrary, precisely because its boundaries are clear and its structure explicit, ASRI possesses the potential for further calibration, revision, and expansion. Methodologically speaking, the value of a framework depends not only on whether it is perfect, but more on whether it can advance the issue to a higher-resolution level of discussion. On this point, ASRI V1.0 has already accomplished its most critical task: it has enabled the question of "Will AI replace a certain occupation?" to no longer remain solely within the realm of vague intuition and positional debate, but to begin to possess the possibility of structured analysis.

8. Implications for Subsequent Research

What ASRI opens up is not only an indicator, but also a research pathway. Future research can unfold along multiple directions.

First, establish a multi-rater mechanism, implementing dual-scorer or multi-scorer independent scoring for the same occupational scenario, and use statistical consistency methods to test inter-rater reliability. Second, use the Delphi method to calibrate anchors, inviting participants with relevant occupational research or industry experience to provide multiple rounds of anonymous feedback and aggregation on boundary cases. Third, establish a standardized case database covering key sectors such as education, healthcare, law, manufacturing, content industry, and platform labor, providing a foundation for horizontal comparison and scoring training. Fourth, explore the correlation between ASRI and external real-world indicators such as changes in job demand, recruitment structure, and wage stratification to test the boundaries of its explanatory power. Fifth, while keeping the main framework consistent, develop more detailed supplementary anchor descriptions for different fields.

Going further, ASRI may also become an auxiliary language for understanding occupational policy, educational training, and capability transformation. If it is possible to identify with relative stability which task parts within a certain type of occupation are more likely to be compressed and which defensive capabilities are more likely to be elevated, then vocational education, job training, and organizational transformation strategies can move beyond abstract slogans like "learn AI/don't use AI" and shift toward more concrete task capability reconstruction.

9. Summary

Overall, the discussion value of ASRI V1.0 lies in the fact that it neither exaggerates the impact of artificial intelligence on occupations into a simple total substitution, nor uses the banner of "human uniqueness" to obscure the structural pressures in reality. It attempts to

establish a more analyzable middle ground between the two: on one hand, acknowledging that many occupational tasks are becoming more susceptible to AI takeover, embedding, and restructuring; on the other hand, also identifying those human task structures that still constitute the defensive core of occupations.

Therefore, the true significance of ASRI does not lie in providing a final answer, but in providing a structured ruler. With this ruler, we need no longer only ask "who will disappear," but can further inquire: which tasks will change first, which capabilities will be revalued, which positions will be compressed, which responsibilities will be re-centralized, and are occupations ultimately heading toward substitution or toward reshaping?

Conclusion

Centered on the core issue of occupational change in the era of artificial intelligence, this paper has proposed the AI Substitution and Reshaping Index (ASRI) V1.0, aiming to provide a more fine-grained and structurally explanatory methodological framework for the public discussion on "whether occupations will be replaced by AI." Distinct from traditional judgment methods that take the occupational title as the basic unit, the core stance of ASRI is that the impact of artificial intelligence on the world of work first occurs at the task level, not the occupational label level. Therefore, what truly needs to be analyzed is not the abstract occupation itself, but the **task composition structure of a specific occupation within a specific scenario**.

Based on this methodological stance, this paper has constructed the six-dimensional structural framework of ASRI V1.0 and grouped it into two mutually countervailing forces: the Degree of Proceduralization, Degree of Digitalization, and Verifiability of Outcomes constitute the substitution dimension group; Ambiguous Adaptability, Interpersonal Empathy, and Non-standard Operations constitute the defensive dimension group. On this basis, ASRI organizes the substitutable parts and the defensive parts within the occupational task structure into a net structural substitution pressure indicator, in order to characterize the structural direction—whether substitution, embedding, restructuring, or reshaping—that an occupation is more likely to experience under the conditions of artificial intelligence.

An important conclusion of this paper is that the discussion on the relationship between occupations and artificial intelligence should not remain stuck on the binary question of "will they be replaced or not?" In reality, occupational change more often manifests as the partial automation of internal tasks, functional migration, job rearrangement, and capability differentiation, rather than the synchronous disappearance of entire occupations. Precisely for this reason, ASRI is better suited to explain which tasks within an occupation are more likely to change first, which parts still constitute the defensive core, and why significant stratification may emerge within the same occupation, rather than serving as an unemployment rate predictor, an elimination probability model, or a substitution timeline.

This paper simultaneously emphasizes that the proposal of ASRI does not constitute a deterministic judgment on the future of occupations. What ASRI measures is the structural substitution pressure at the task structure level, not the inevitably occurring employment outcomes in the real world. Occupational change will still be profoundly influenced by external factors such as institutional arrangements, liability attribution, legal regulation, organizational adoption, social trust, and cost-benefit considerations. Therefore, ASRI should be understood as an analytical ruler used to characterize structural pressure, rather than a final model for directly predicting real-world fates.

As a V1.0 version, ASRI has already formed a relatively complete conceptual definition, six-dimensional framework, dual-group logic, formula system, and scoring method foundation. However, several explicit limitations still exist, including the persistence of scoring subjectivity, the not yet systematically tested inter-rater reliability, the relatively high demands of continuous scoring on anchor training, the need for further expansion of the case database, and the fact that institutional implementation layer variables have not yet been independently incorporated into the model. It is also for these reasons that ASRI V1.0 should be understood more as a methodological starting point that already possesses use-value but still requires continuous calibration and empirical verification.

The main contribution of this paper lies in advancing the originally highly generalized and often impressionistic debate of "Will AI replace a certain occupation?" into a structural analysis framework that is decomposable, interpretable, recordable, comparable, reviewable, and continuously revisable. It proposes a new observational pathway: rather than viewing occupations as static labels, it is better to understand them as dynamic structures composed of different tasks; rather than simply judging whether an occupation is safe, it is better to analyze how its internal substitution factors and defensive factors interact and shape the direction of change.

Overall, the significance of ASRI V1.0 does not lie in providing a final answer regarding the future of all occupations, but in offering a preliminary yet usable analytical language for occupational research in the era of artificial intelligence. It provides a methodological foundation for subsequent occupational case accumulation, scoring consistency calibration, and cross-occupational comparison, and also offers a new theoretical entry point for understanding occupational reshaping, job differentiation, and capability reconstruction. In the future, with the expansion of the case database, the optimization of the scoring system, and the gradual introduction of real-world data, ASRI is expected to develop into a more robust and more explanatory tool for occupational structural analysis.